COPYRIGHT WARNING: Copyright in these original lectures is owned by Monash University. You may transcribe, take notes, download or stream lectures for the purpose of your research and study only. If used for any other purpose, (excluding exceptions in the Copyright Act 1969 (Cth)) the University may take legal action for infringement of copyright.

Do not share, redistribute, or upload the lecture to a third party without a written permission!

FIT3181/5215 Deep Learning

Week 10: Vision Transformer and Model Fine-Tuning Techniques

CE/Lecturer: Dr Trung Le (Clayton)

Lecturer: Dr Binh Nguyen (Clayton)

Lecturer: Dr Ting Fung Fung (Malaysia)

Lecturer: Dr Arghya Pal (Malaysia)

Lecturer: Dr Sailaja Rajanala (Malaysia)

Email: trunglm@monash.edu

GROUP OF EIGHT AUSTRALIA

Department of Data Science and AI
Faculty of Information Technology, Monash University, Australia



Outline

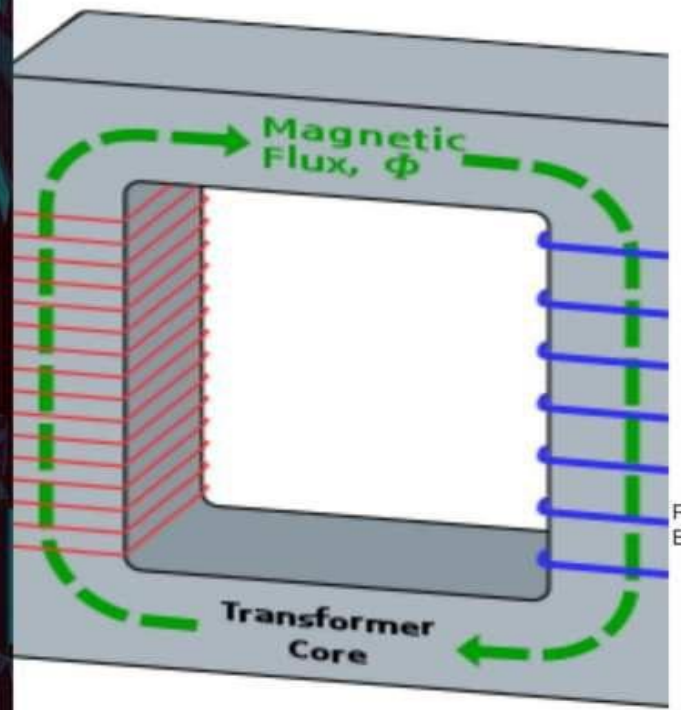
- Revision of Transformers
- Vision Transformer
- Swin Transformer
- Model Fine-Tuning for Transformer
  - Prompt-Tuning
  - LoRA
  - Adapter

- Acknowledgment:
  - https://web.eecs.umich.edu/~justincj/slides/eecs498/WI2022/598_WI2022_lecture18.pdf
  - https://www.slideshare.net/slideshow/transformers-in-vision-from-zero-to-hero-dlipptx/253343336

What do you think when you hear the word «Transformer»?



Transformers at school



Transformers at college

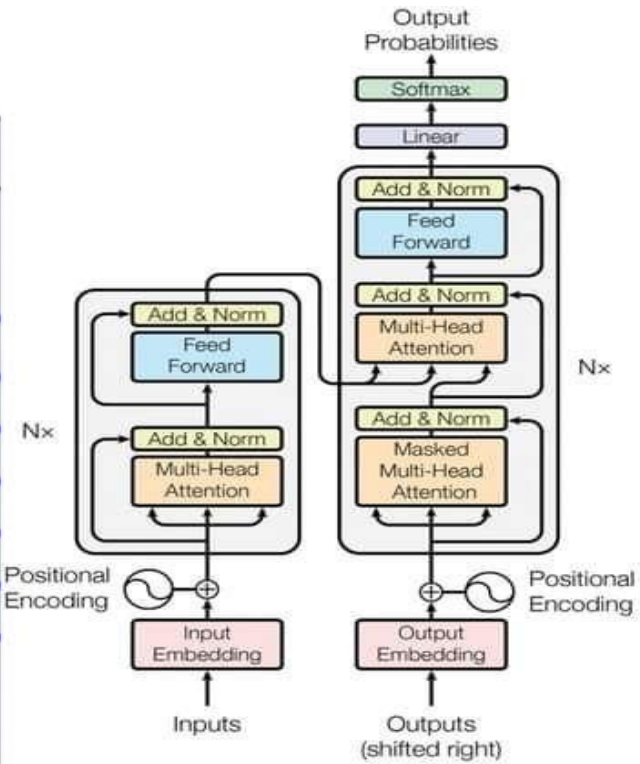
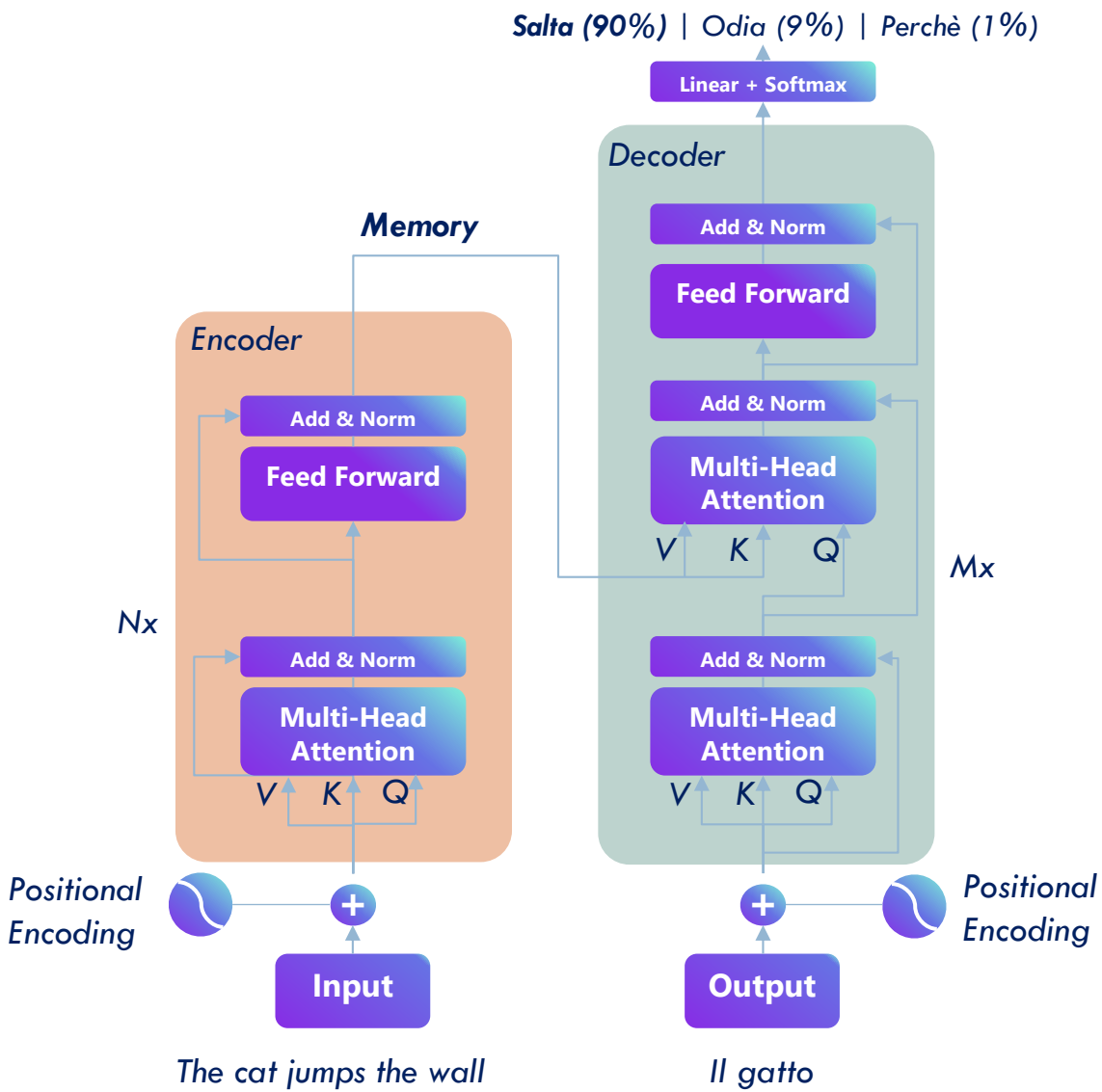
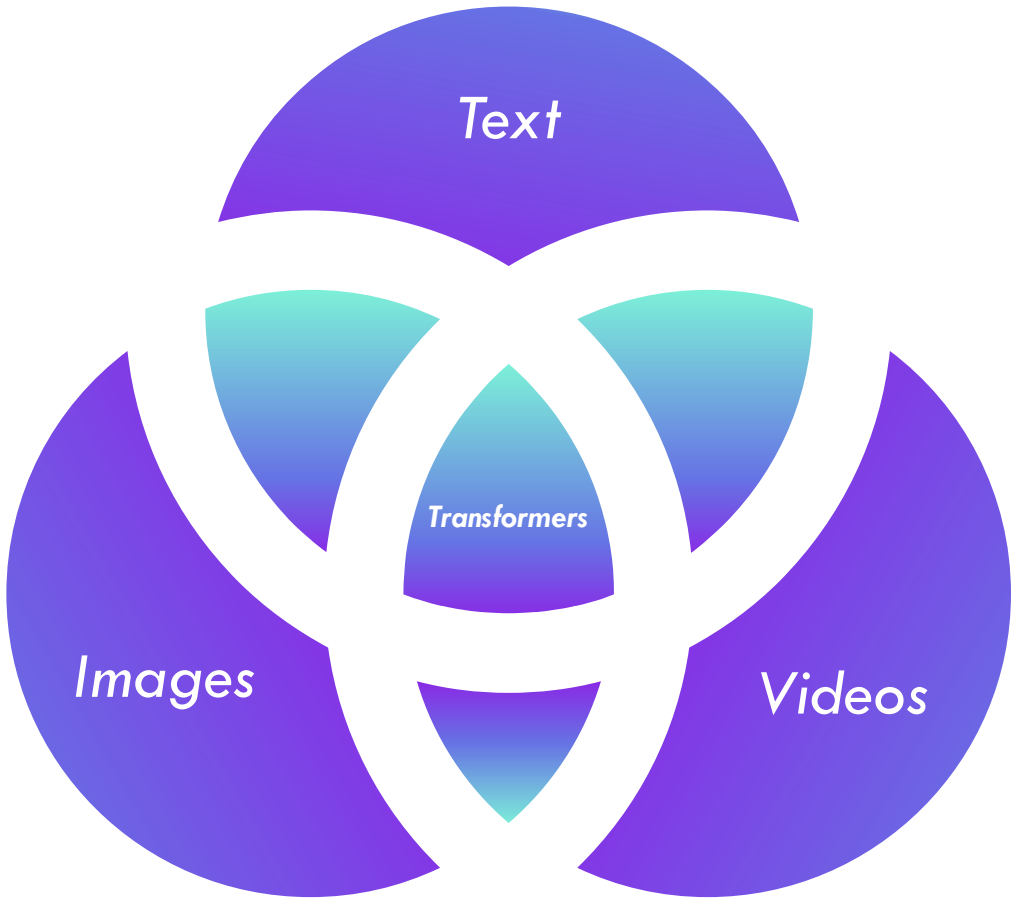


Figure 1: The Transformer - model architecture.

Transformers today

The Transformer «today»





History

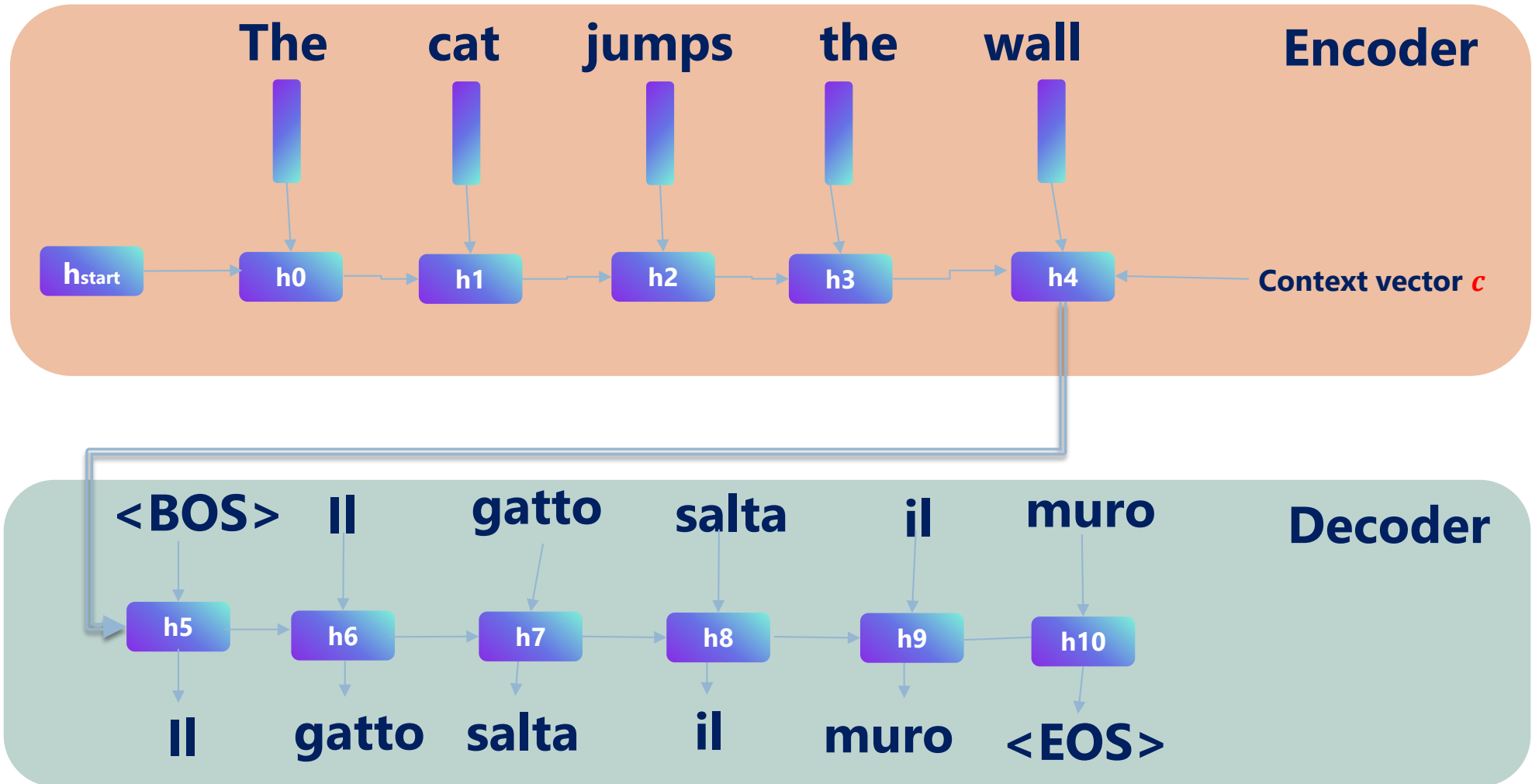
2017 *Introduced transformers in NLP*

2020 *Vision Transformers*

2021 *Transformers for video understanding*

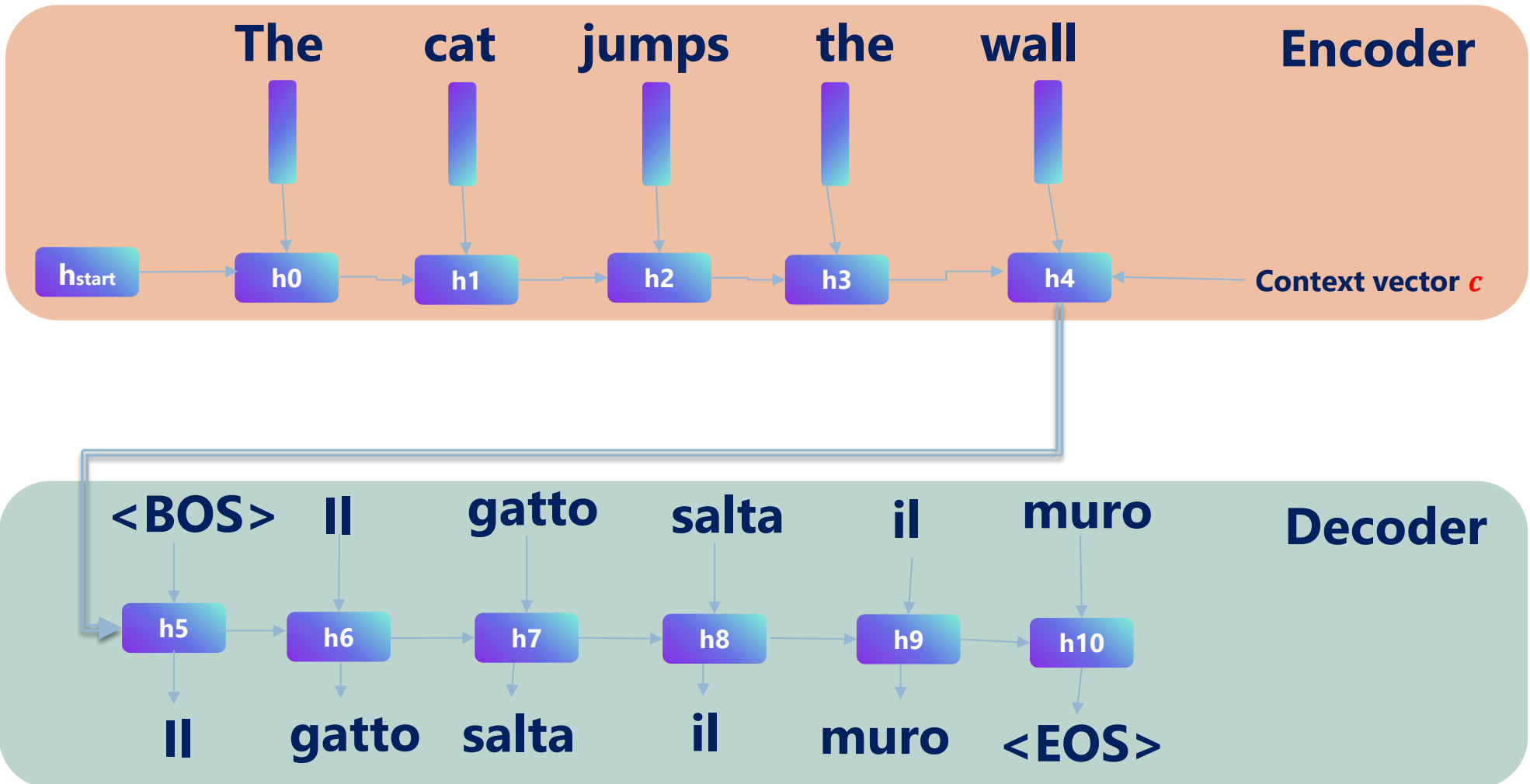
Now *Computer Vision Revolution!*

A step back: Recurrent Networks (RNNs)



Problems

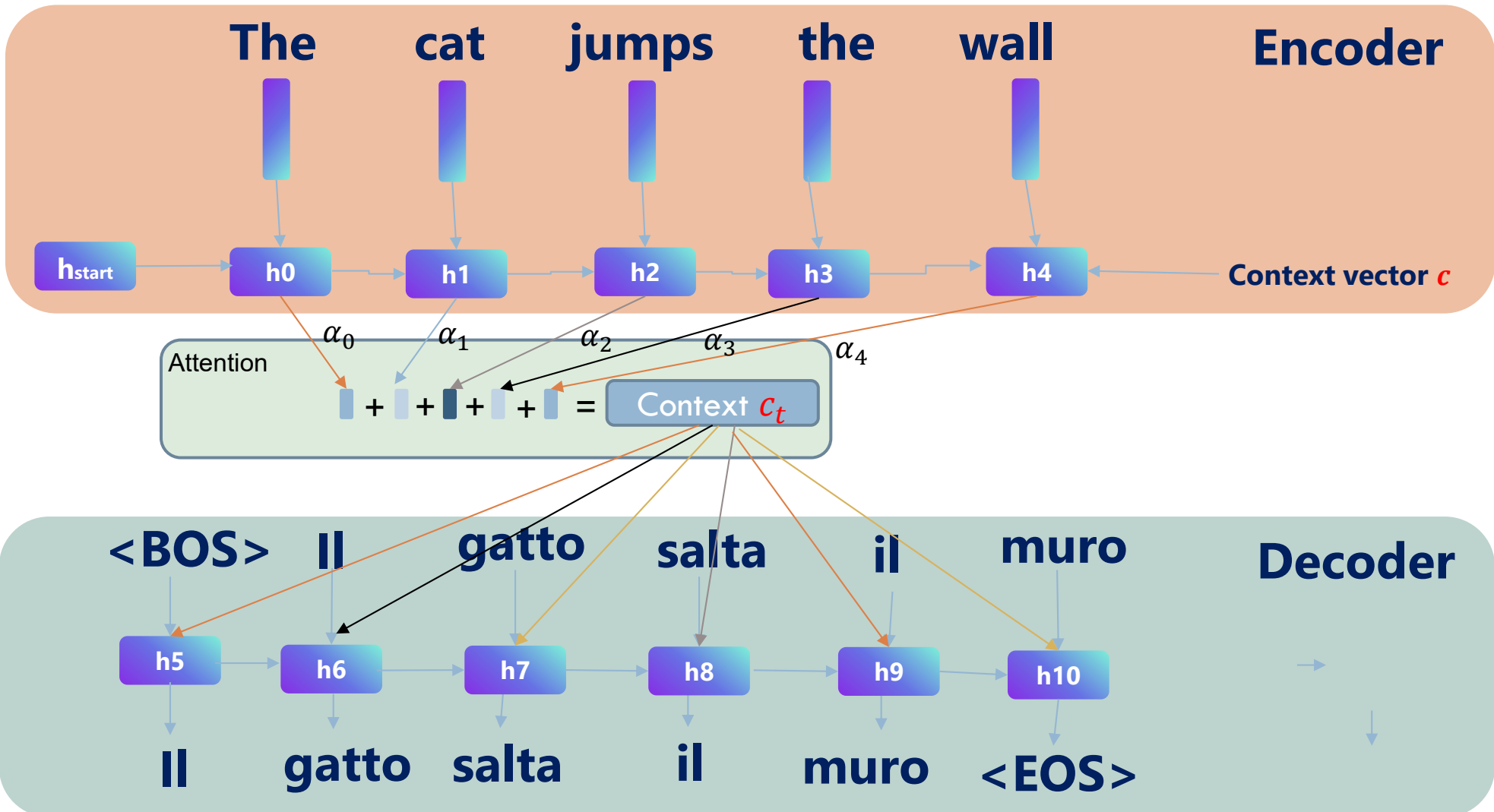
1. We **forget tokens too far** in the past in the context vector **c**
2. We need to wait the previous token to compute the next hidden-state



Solving problem 1

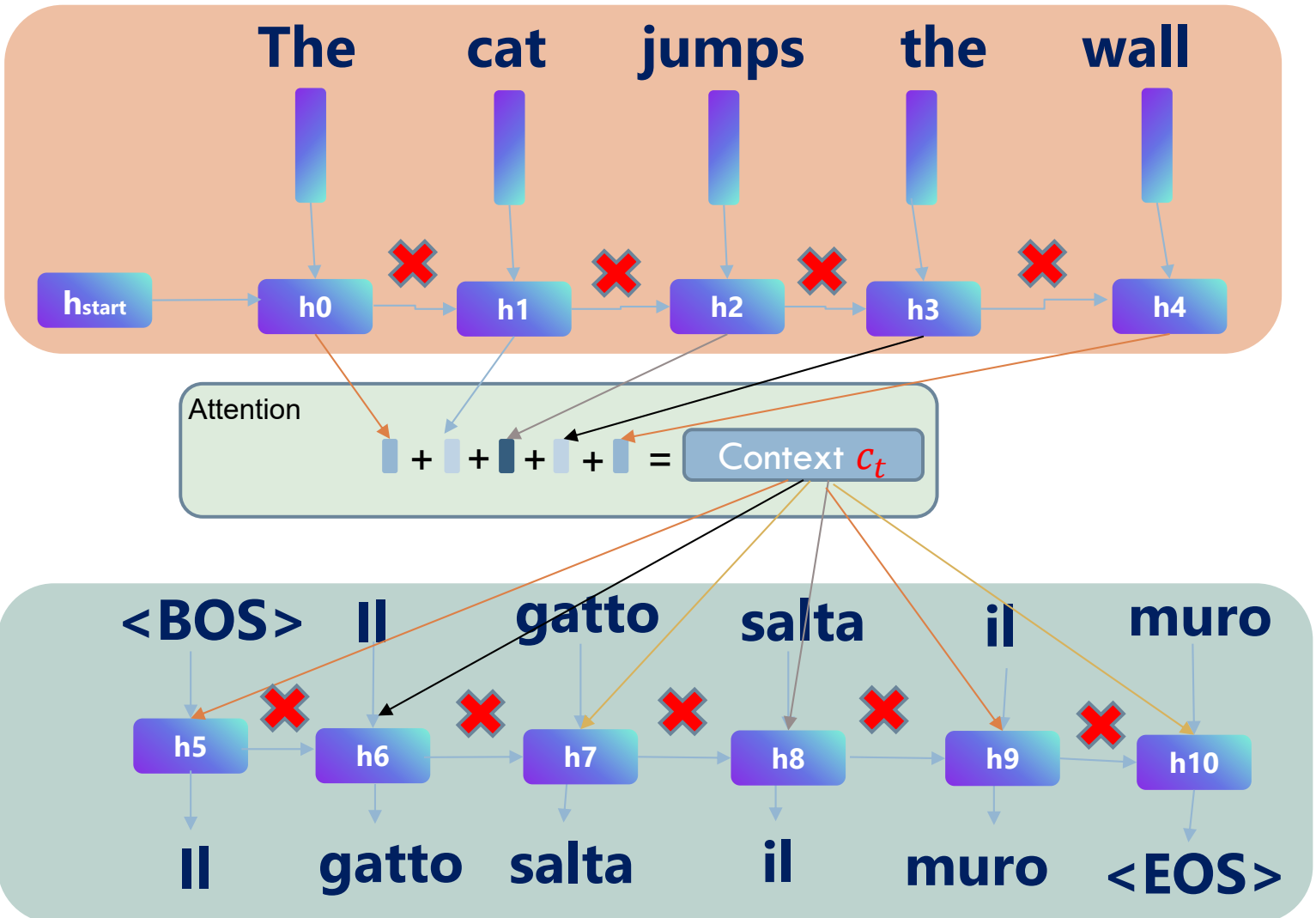
"We **forget tokens too far** in the past"

Solution: Add an **attention** mechanism



Solving problem 2

"We need to wait the previous token to compute the next hidden-state"

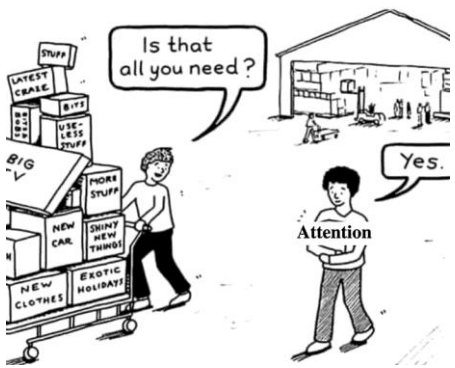


Solution

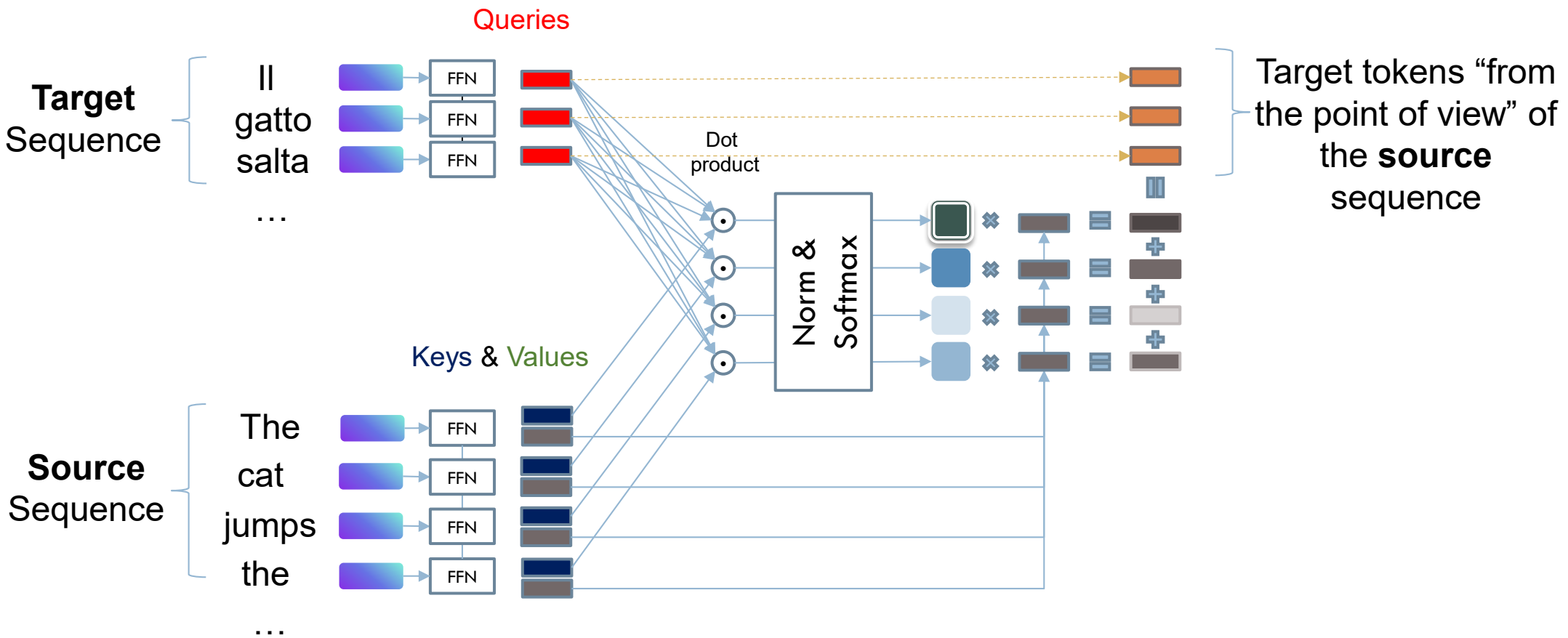
Throw away **recurrent connections**

2017 paper

"Attention Is All You Need"

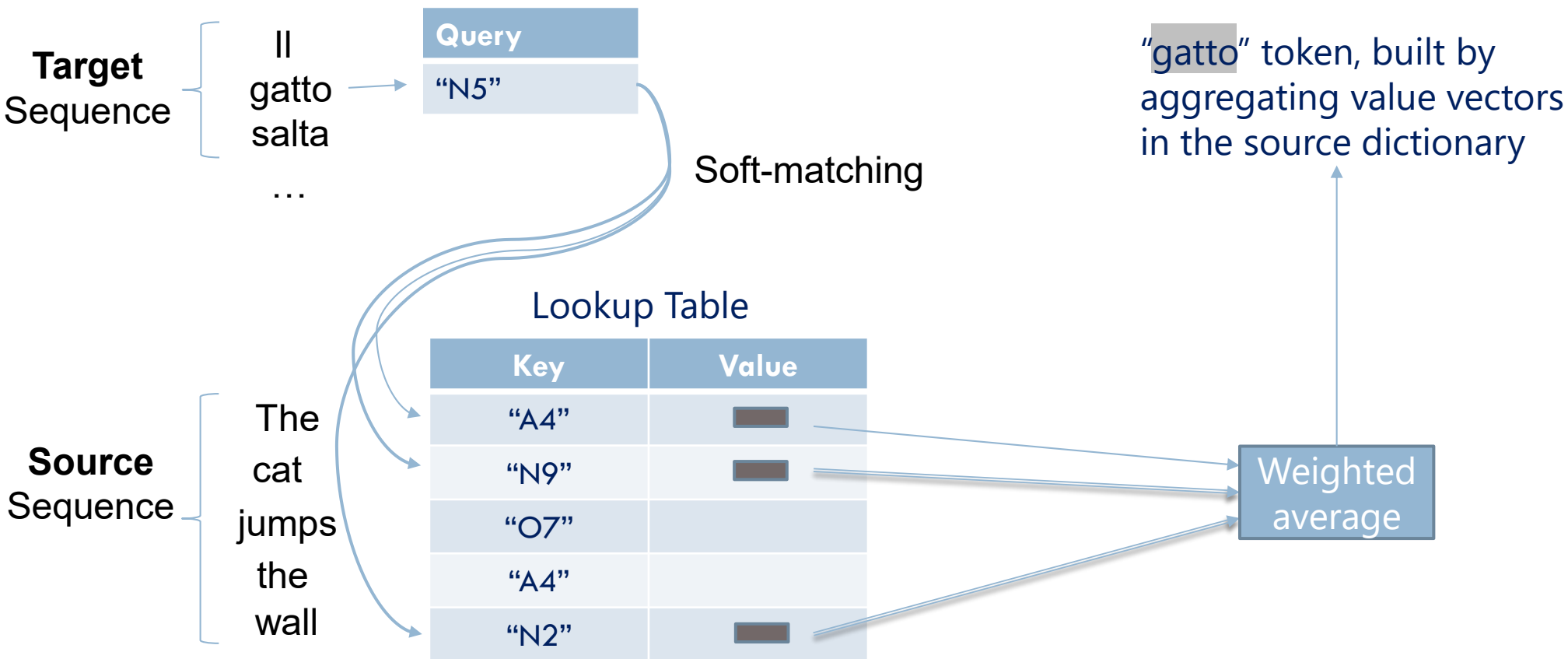


Transformer's Attention Mechanism



Transformer's Attention Mechanism

From a different perspective

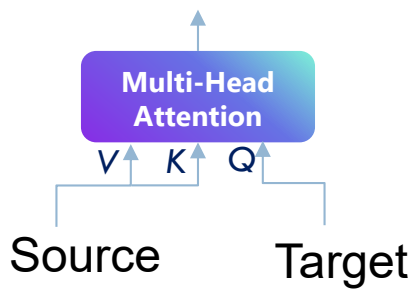


Cross-Attention and Self-Attention

Cross-Attention

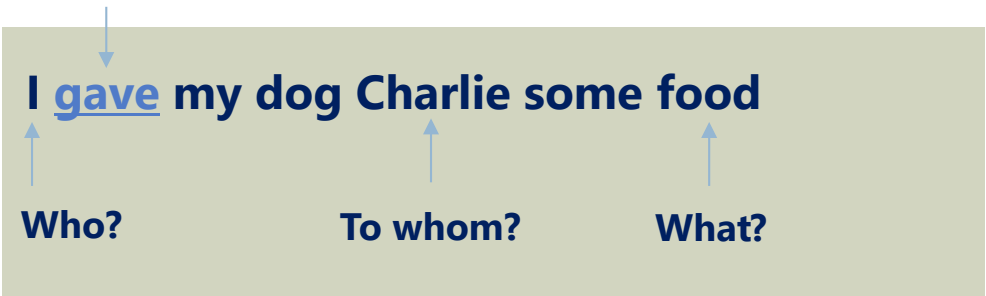


- Source ≠ Target
  - Queries from Target
  - Key, Values from Source



- Captures **inter-**sequence dependencies

Self-Attention



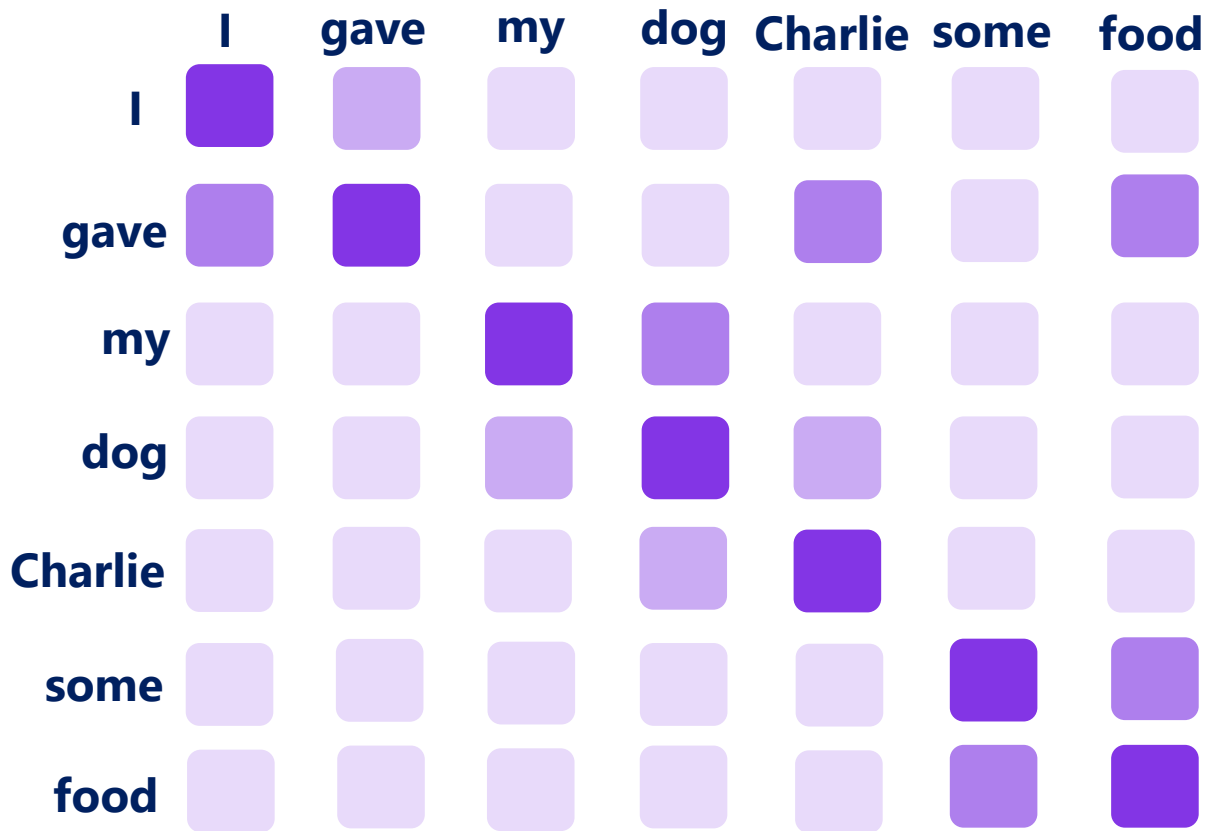
- Source = Target
  - Key, Queries, Values obtained from the same sentence



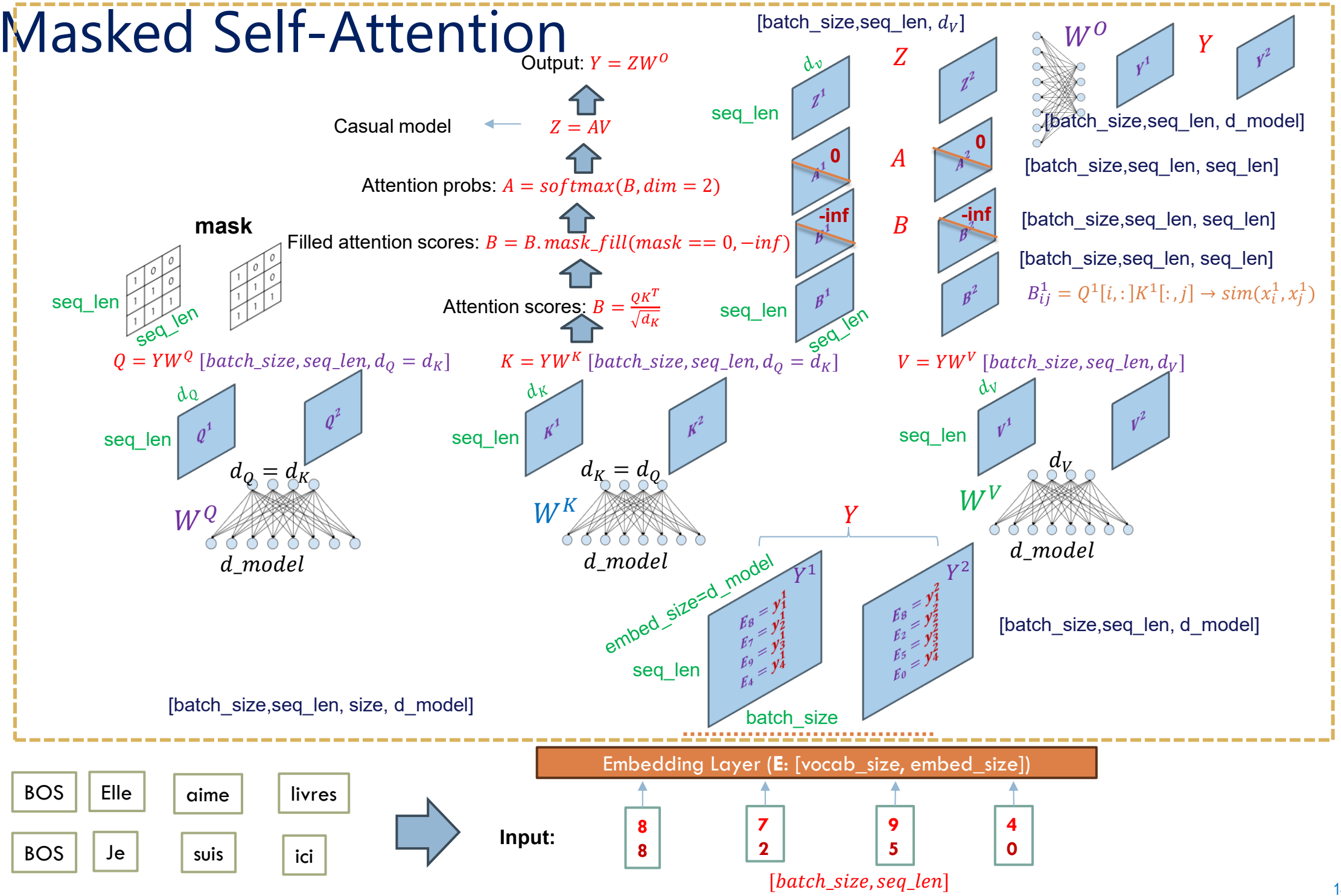
- Captures **intra-**sequence dependencies

Self-Attention

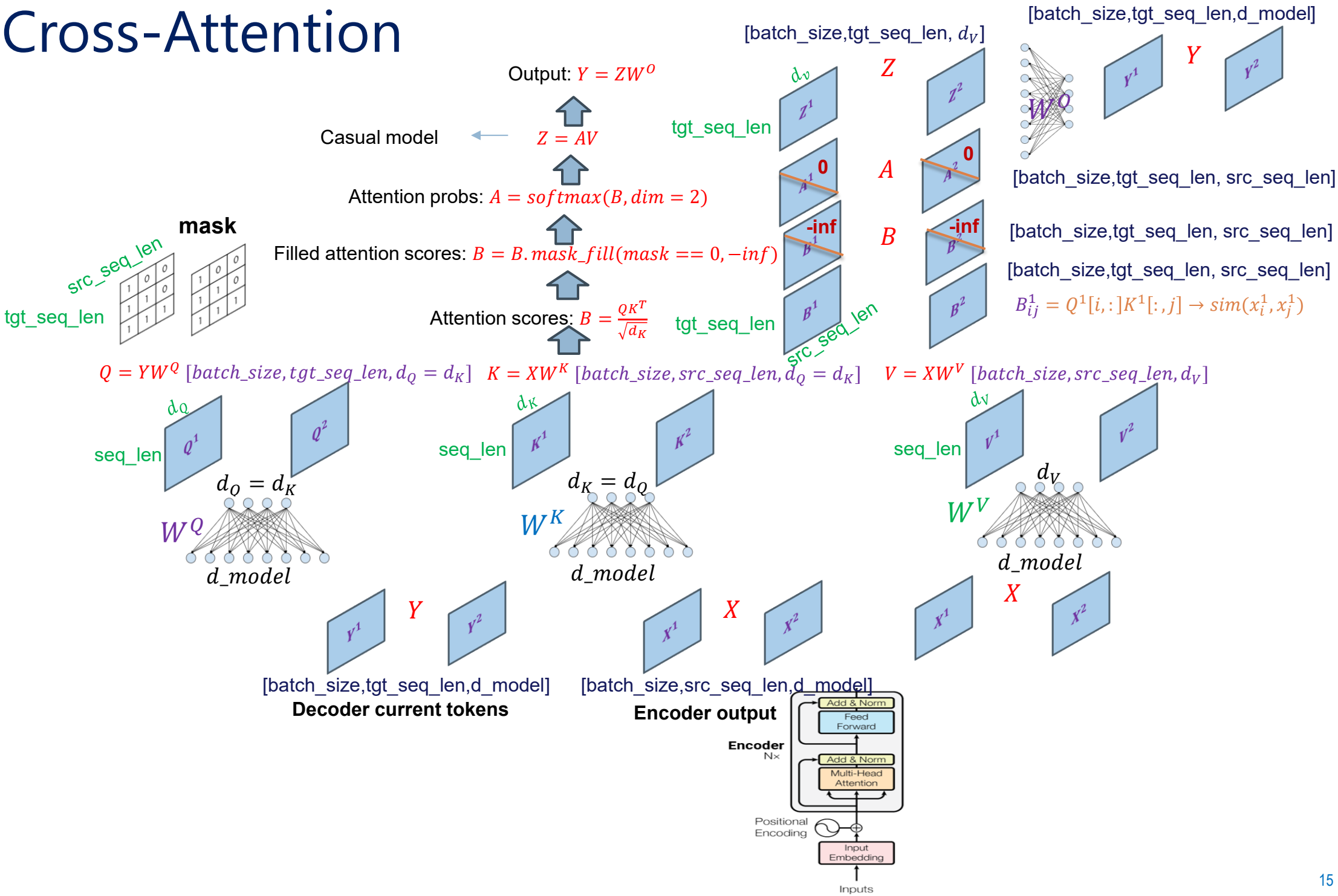
Attention calculation is $O(n^2)$



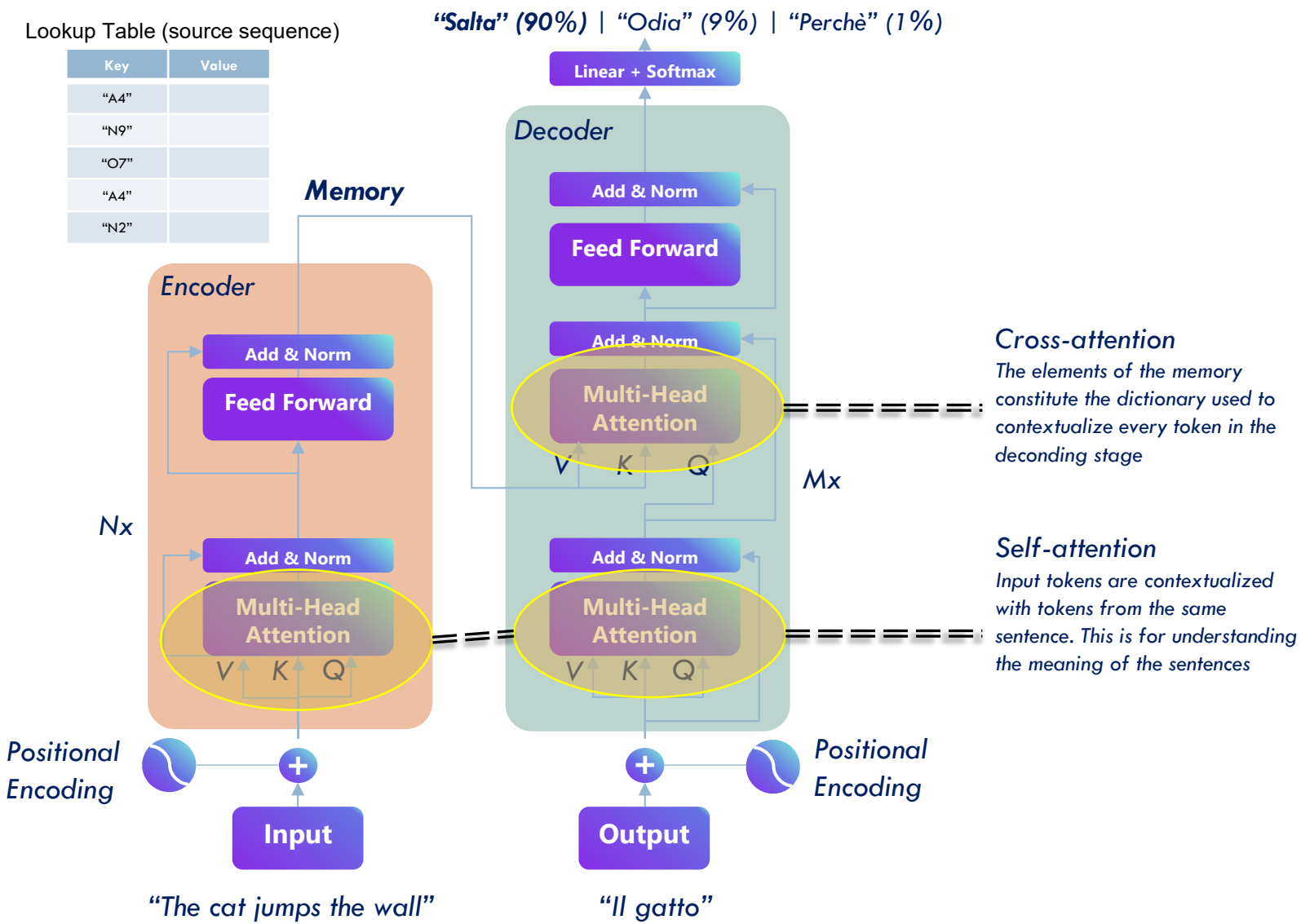
Masked Self-Attention



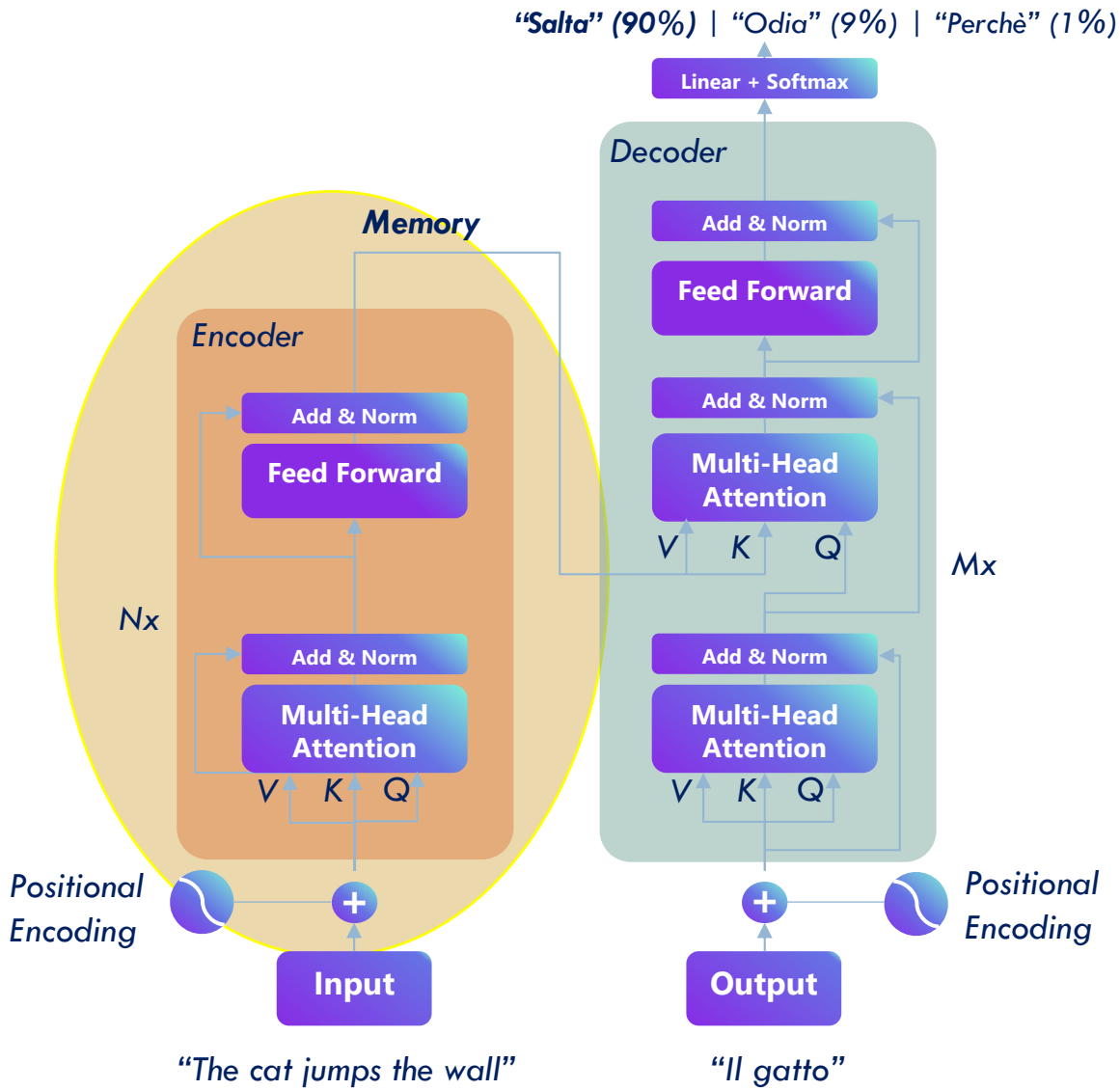
Cross-Attention



Full Transformer Architecture



Full Transformer Architecture



Vision Transformer

Transformers in Computer Vision

Can we use the self-attention mechanism in images?



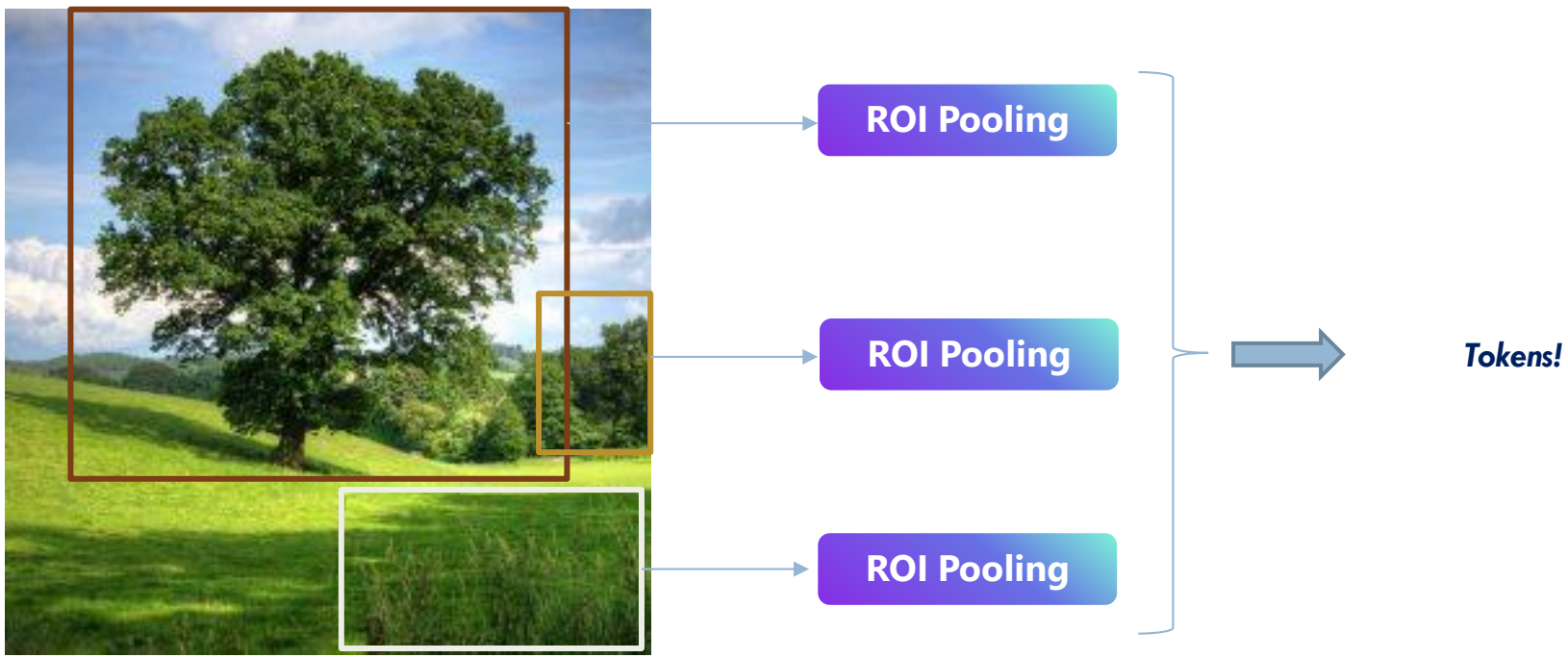
Transformers in Computer Vision

- The transformer works with a set of tokens
- What are **tokens/visual words** in images**?**



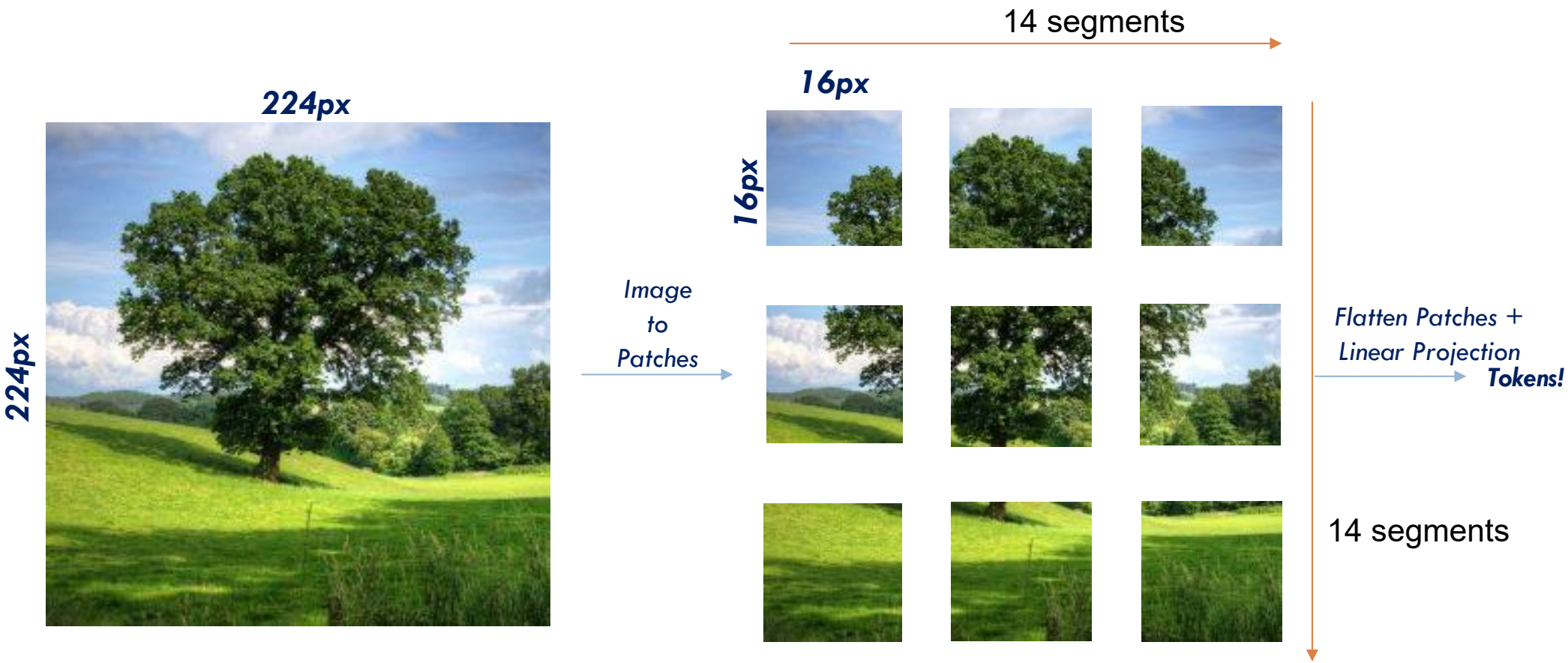
Transformers in Computer Vision

- Tokens as the features from an object detector



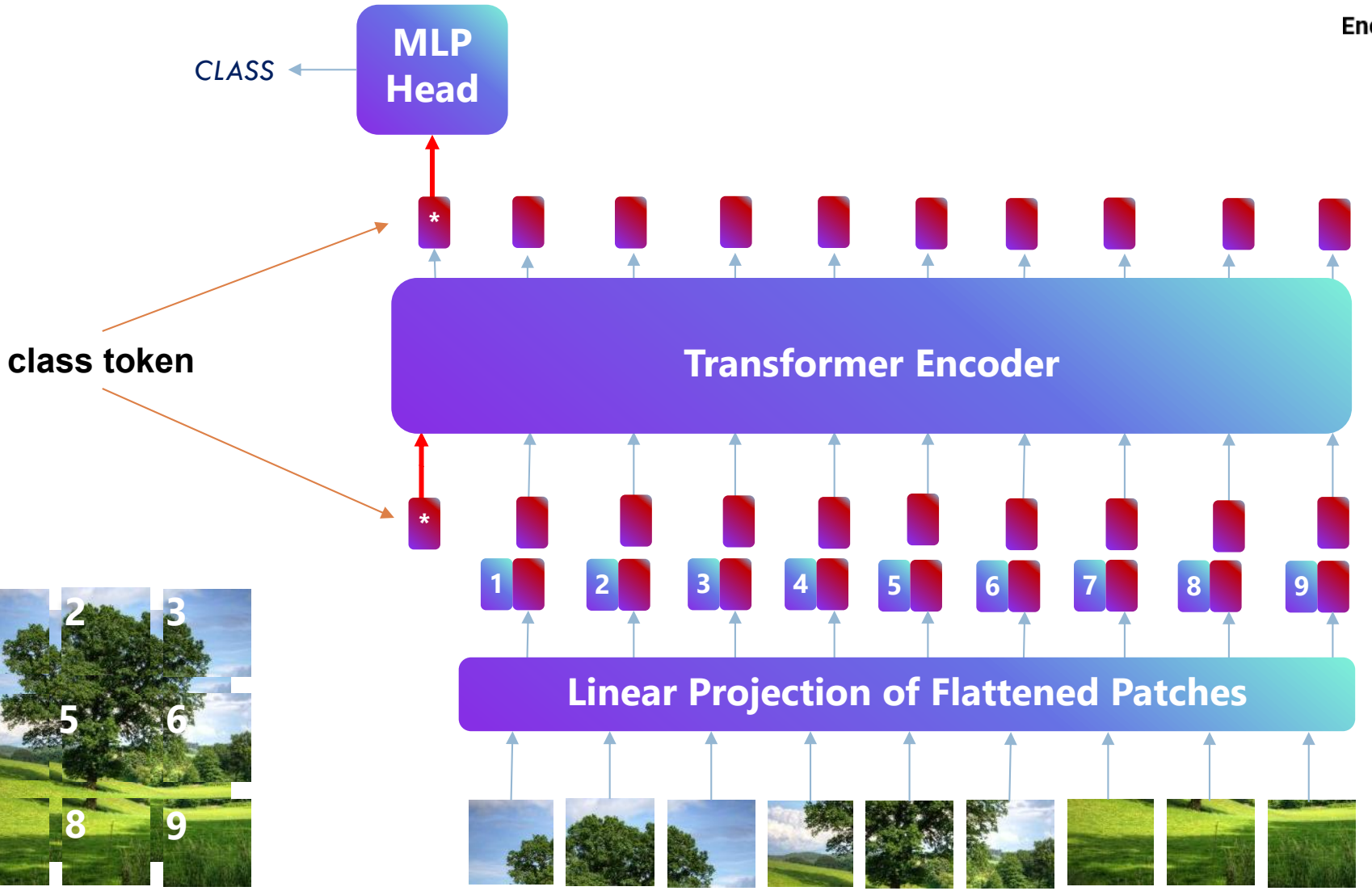
Vision Transformers (ViTs)

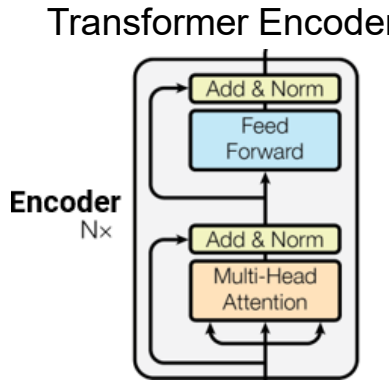
"An image is worth 16x16 words"



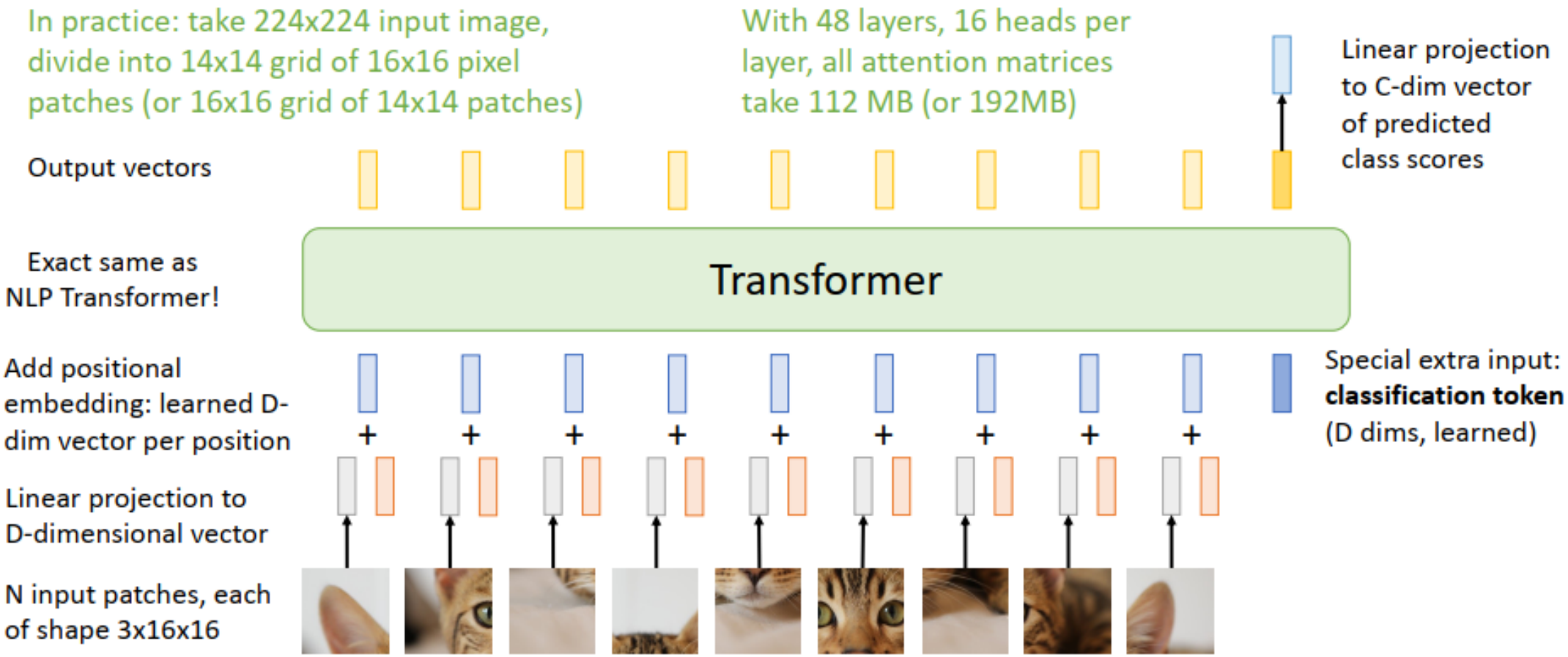
"An image is worth 16x16 words" | *Dosovitskiy et al., 2020*

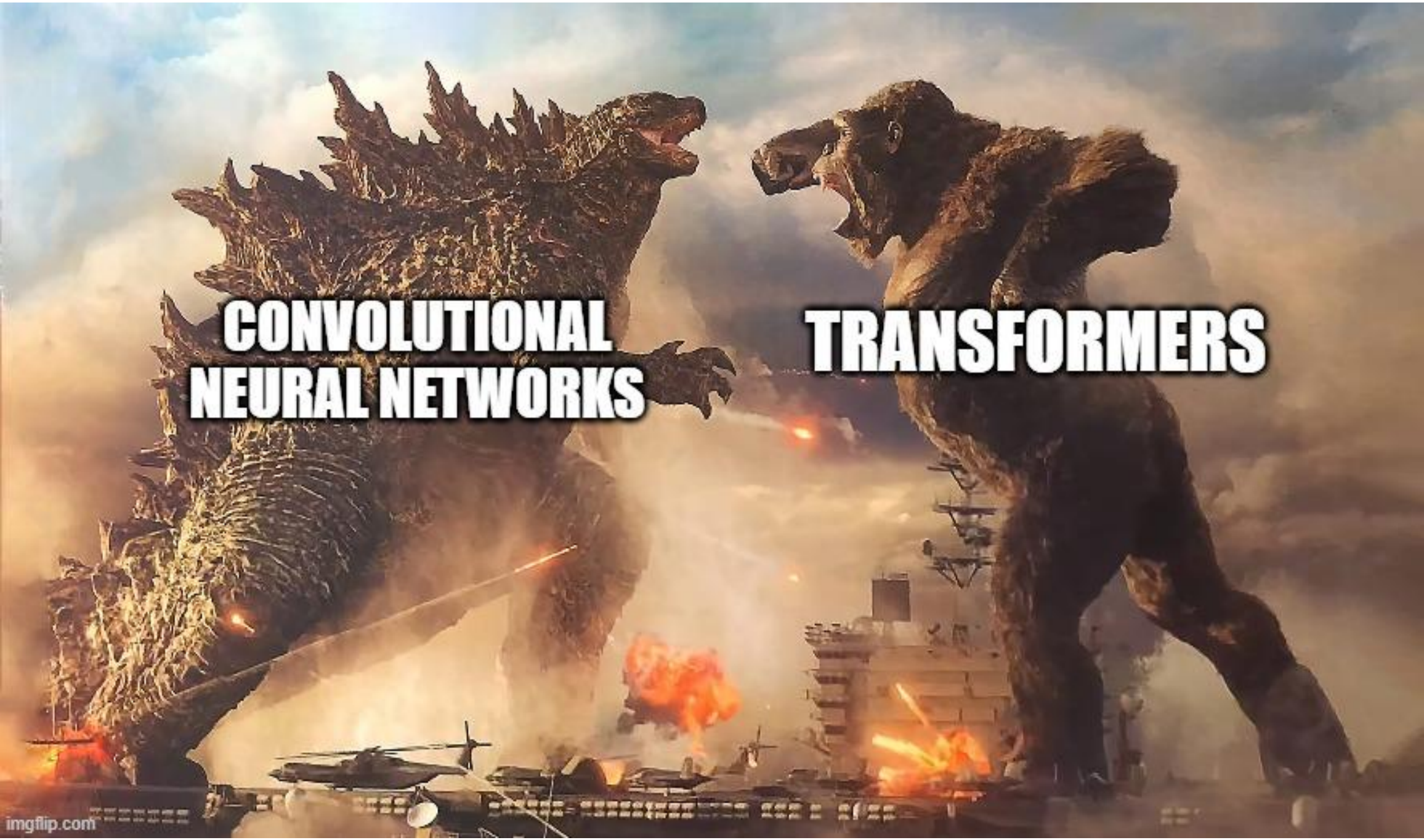
Vision Transformers (ViTs)



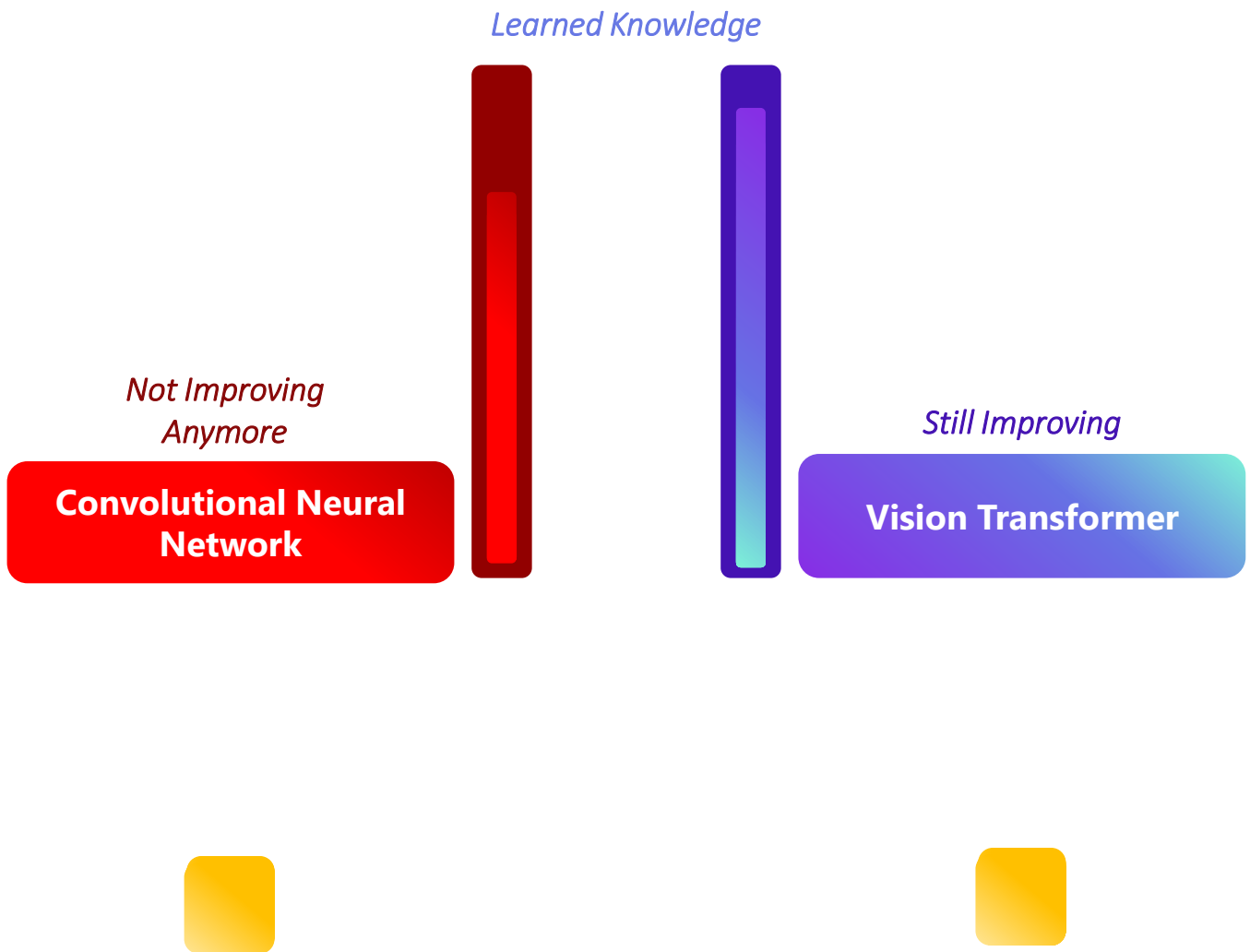


Vision Transformers (ViTs)

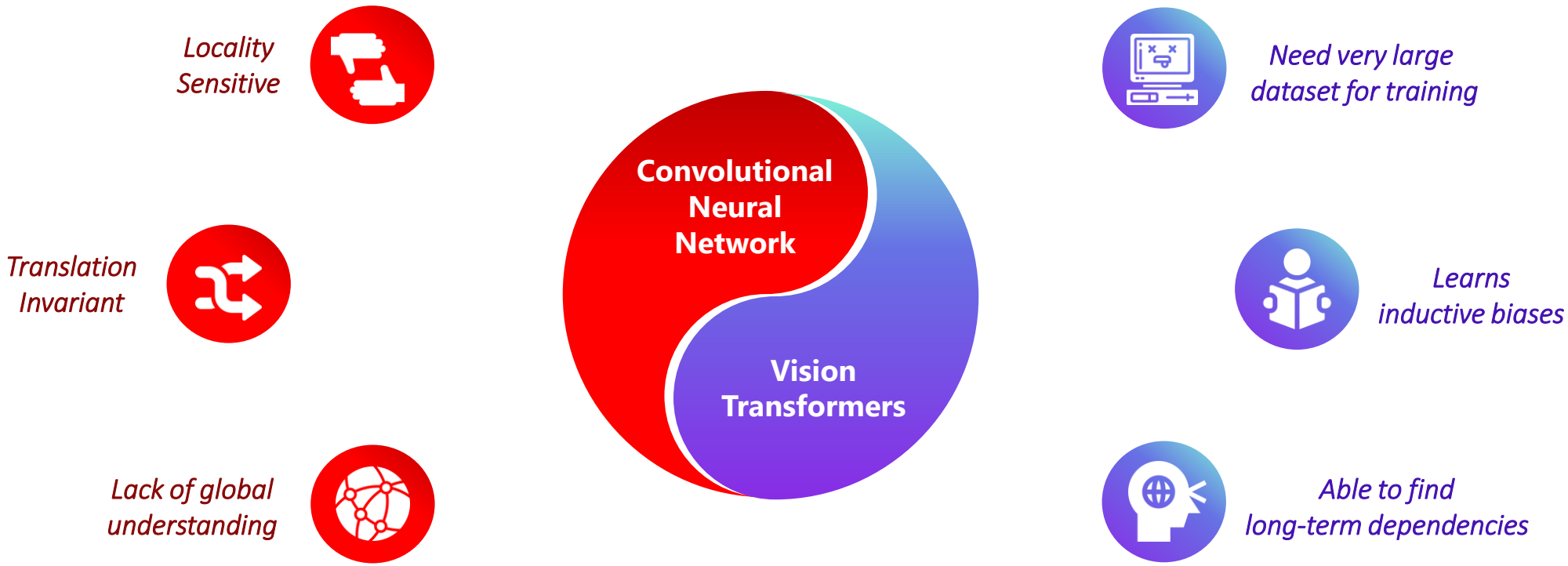



CONVOLUTIONAL
NEURAL NETWORKS
TRANSFORMERS
imgflip.com

What happens during training?

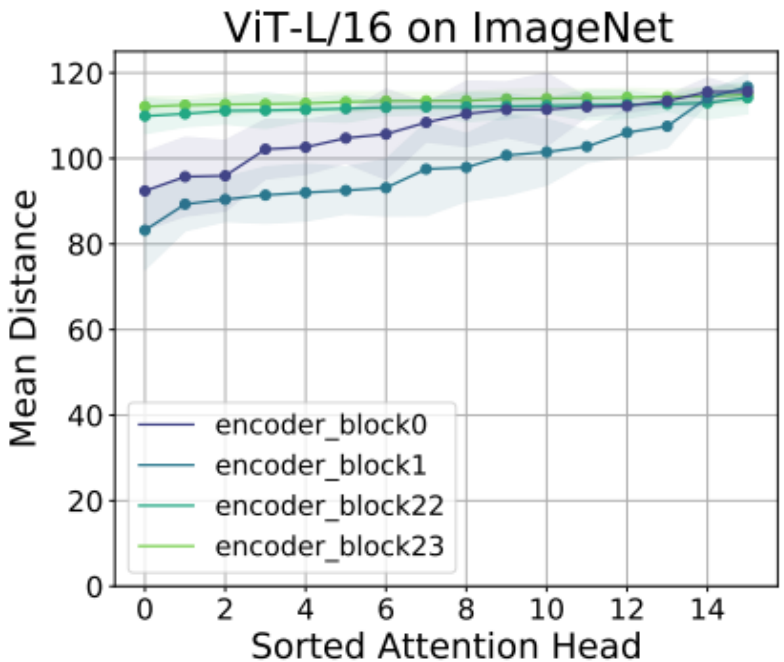


Why are they different?



A different point of view

ViTs are both local and global!



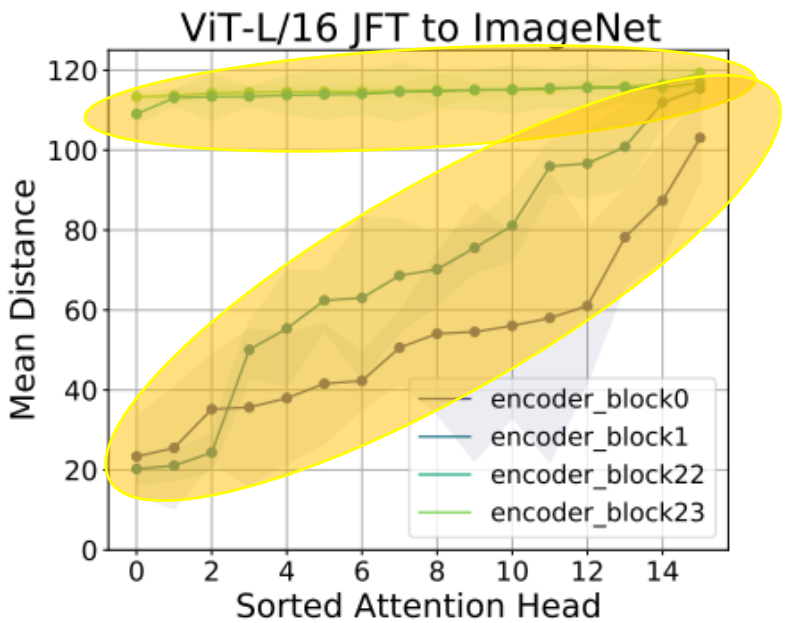
The ViT learns only global information with low amount of data

Heads focus on farther patches



A different point of view

ViTs are both local and global!



The ViT learns also local information with more data

Higher layers heads still focus on farther patches



Lower layers heads focus on both farther and closer patches



A different point of view

Vision Transformers are very robust!

Occlusion



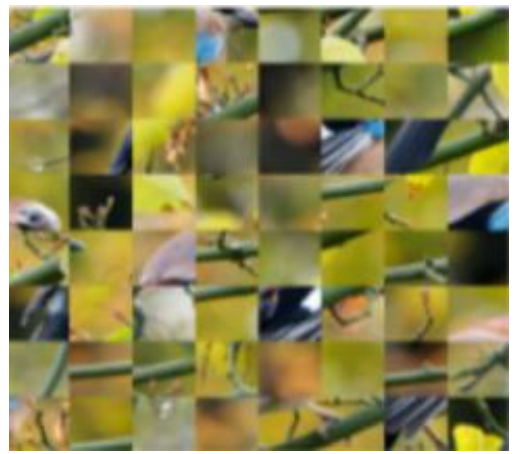
Distribution Shift



Adversarial Perturbation



Permutation

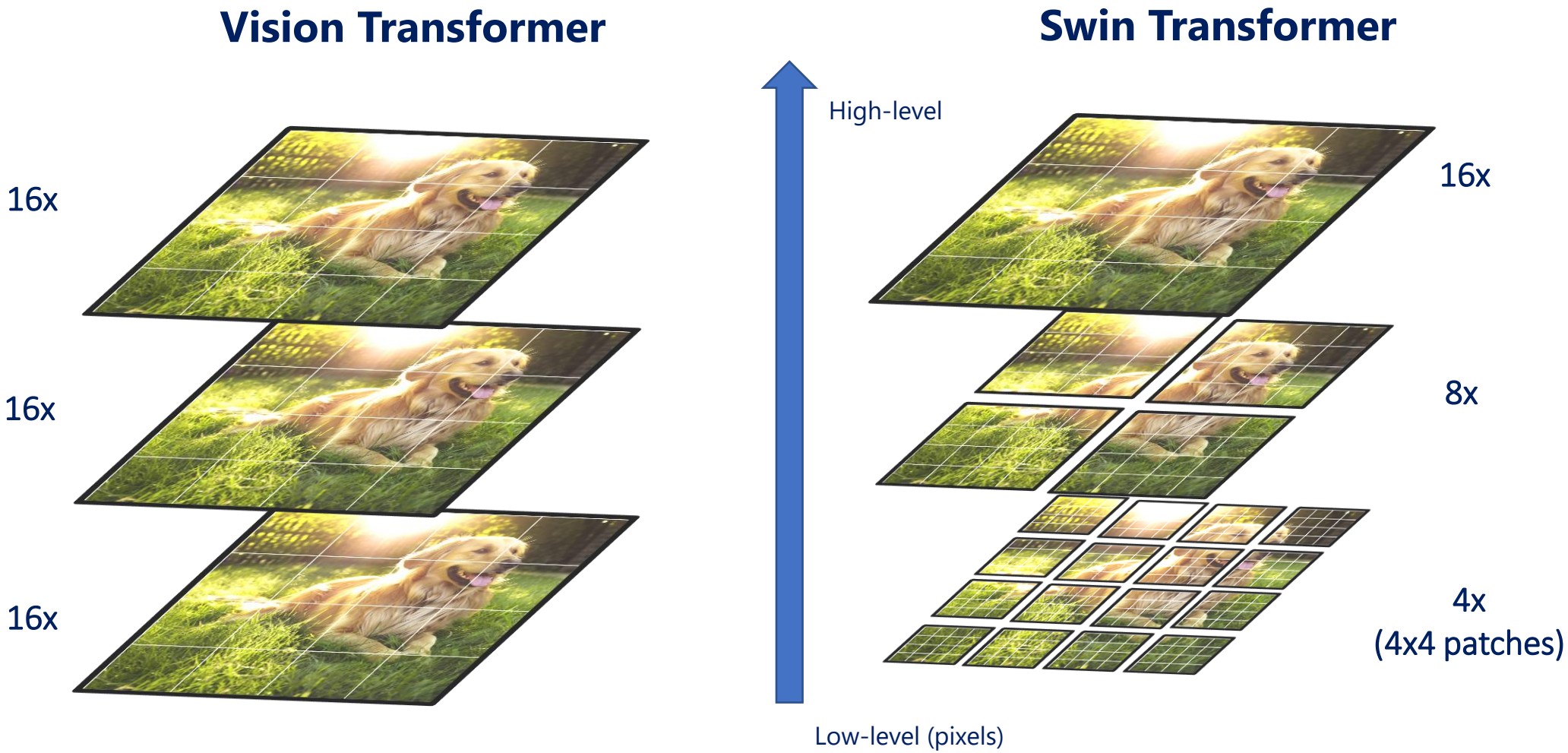


Intriguing Properties of Vision Transformers | Muzammal Naseer et al.

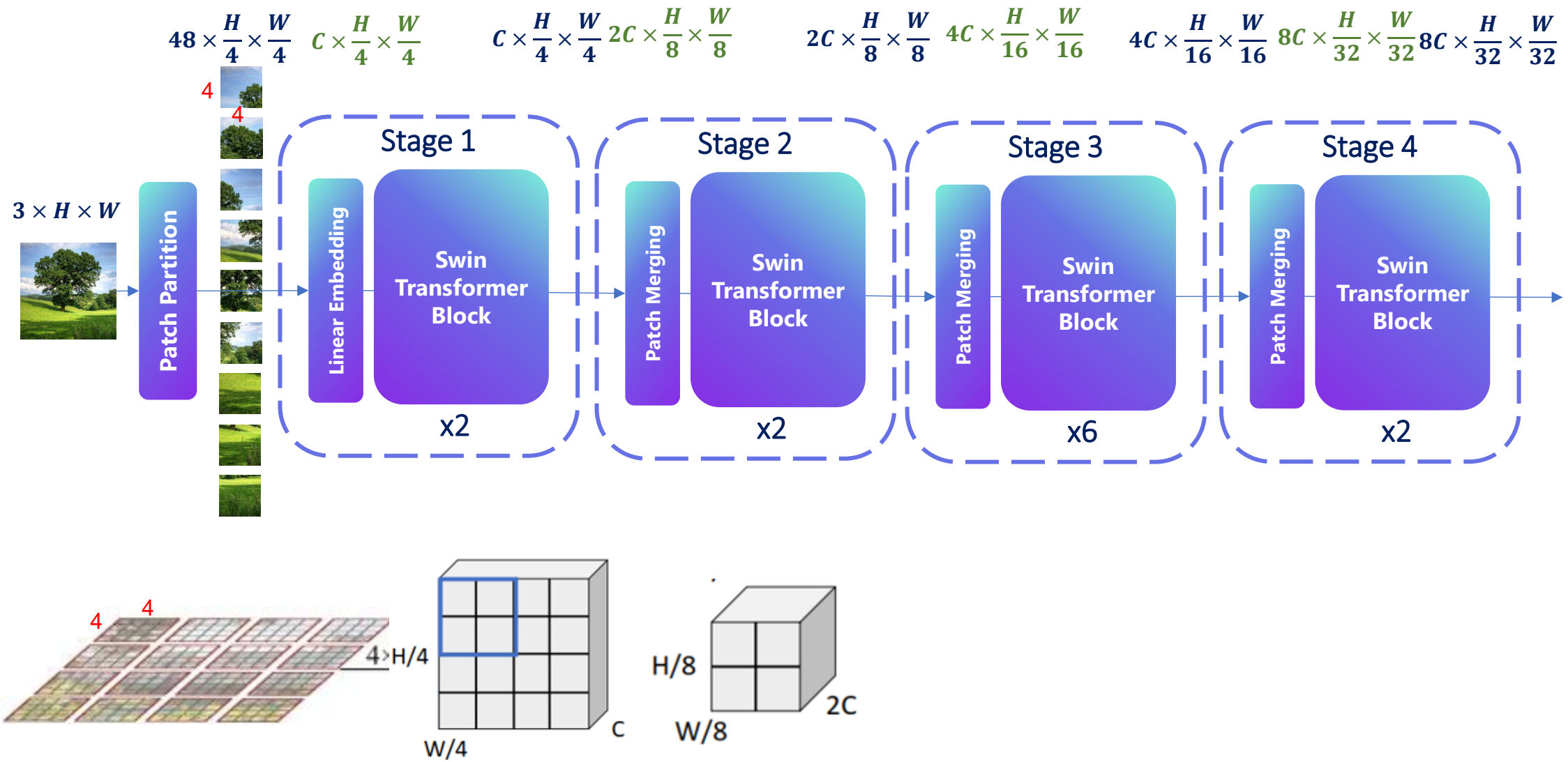
Swin Transformer

Swin Transformers

Shifted Window based Self-Attention

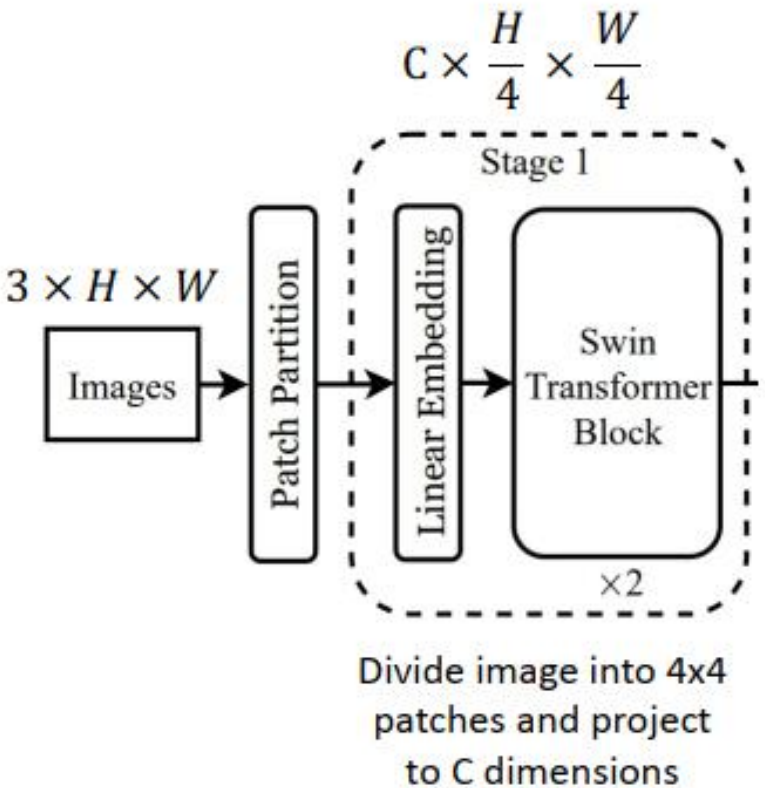


Swin Transformers

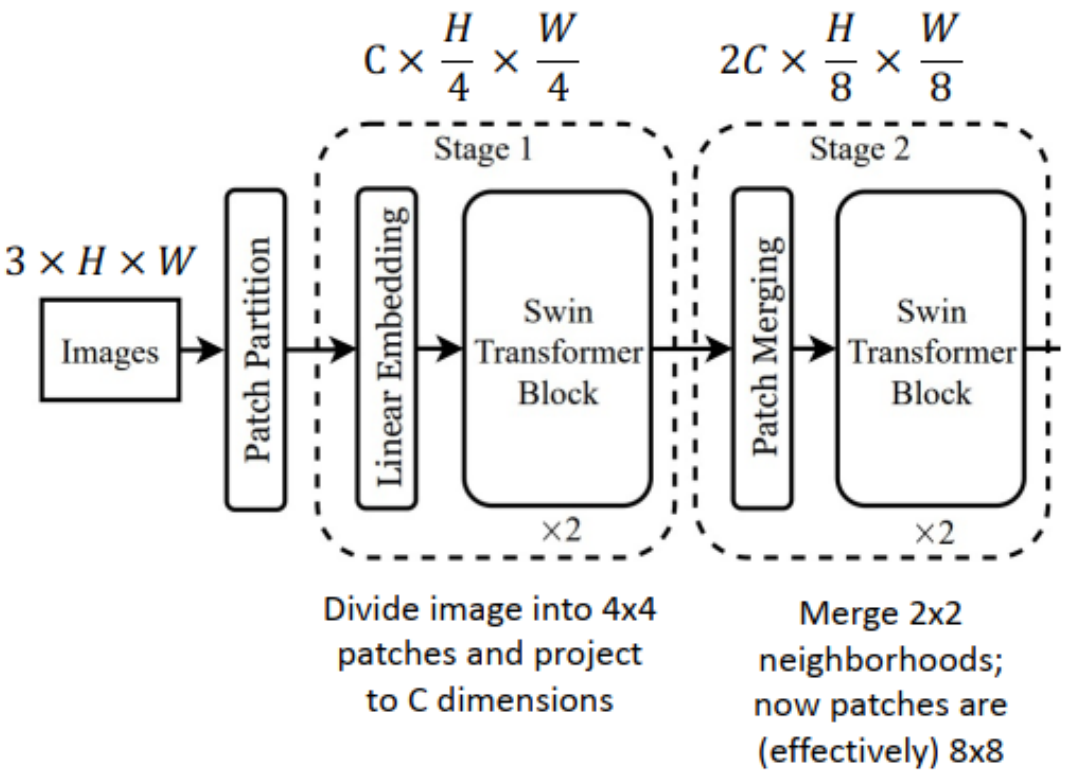


Swin Transformer: Hierarchical Vision Transformer using Shifted Windows | Ze Liu et al., 2021

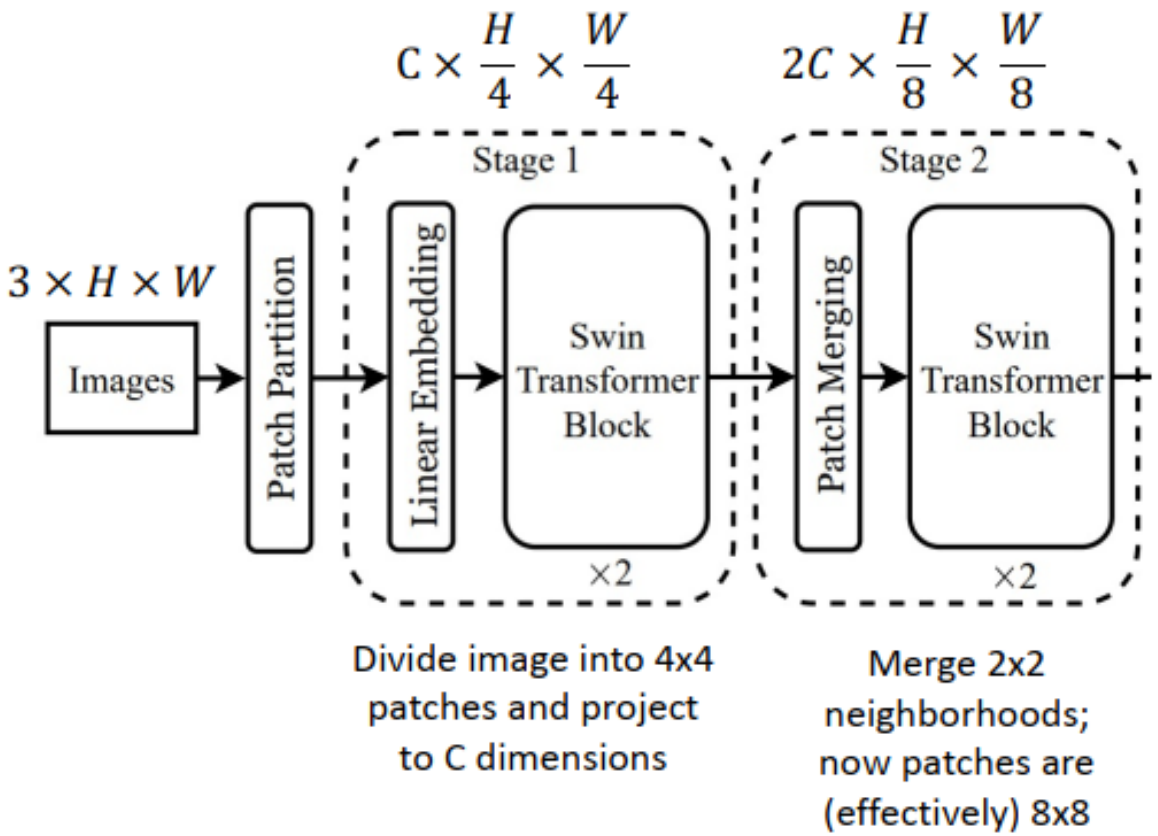
Swin Transformers

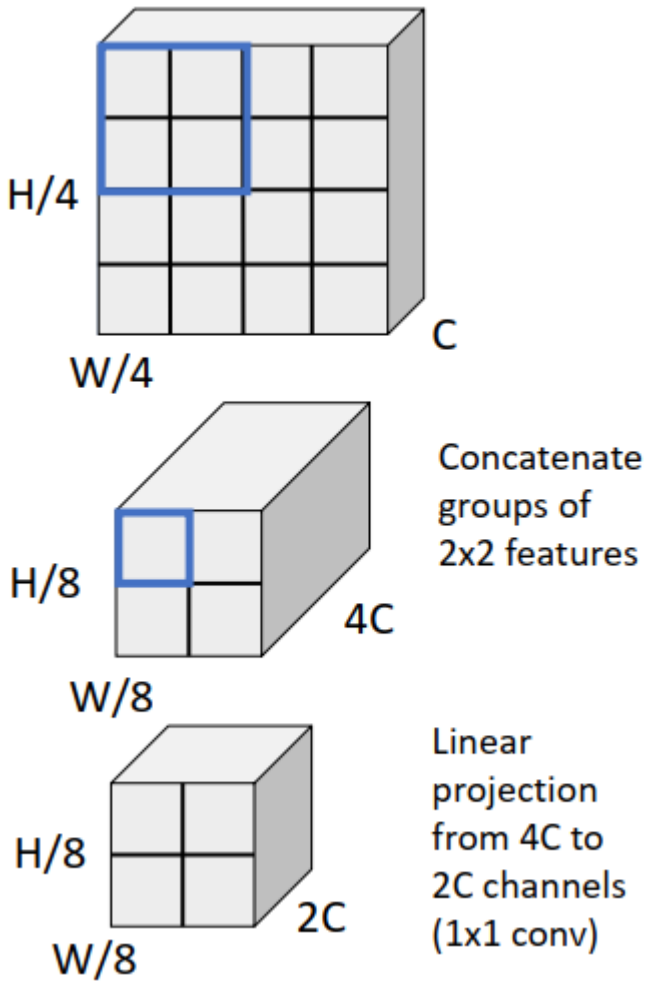


Swin Transformers

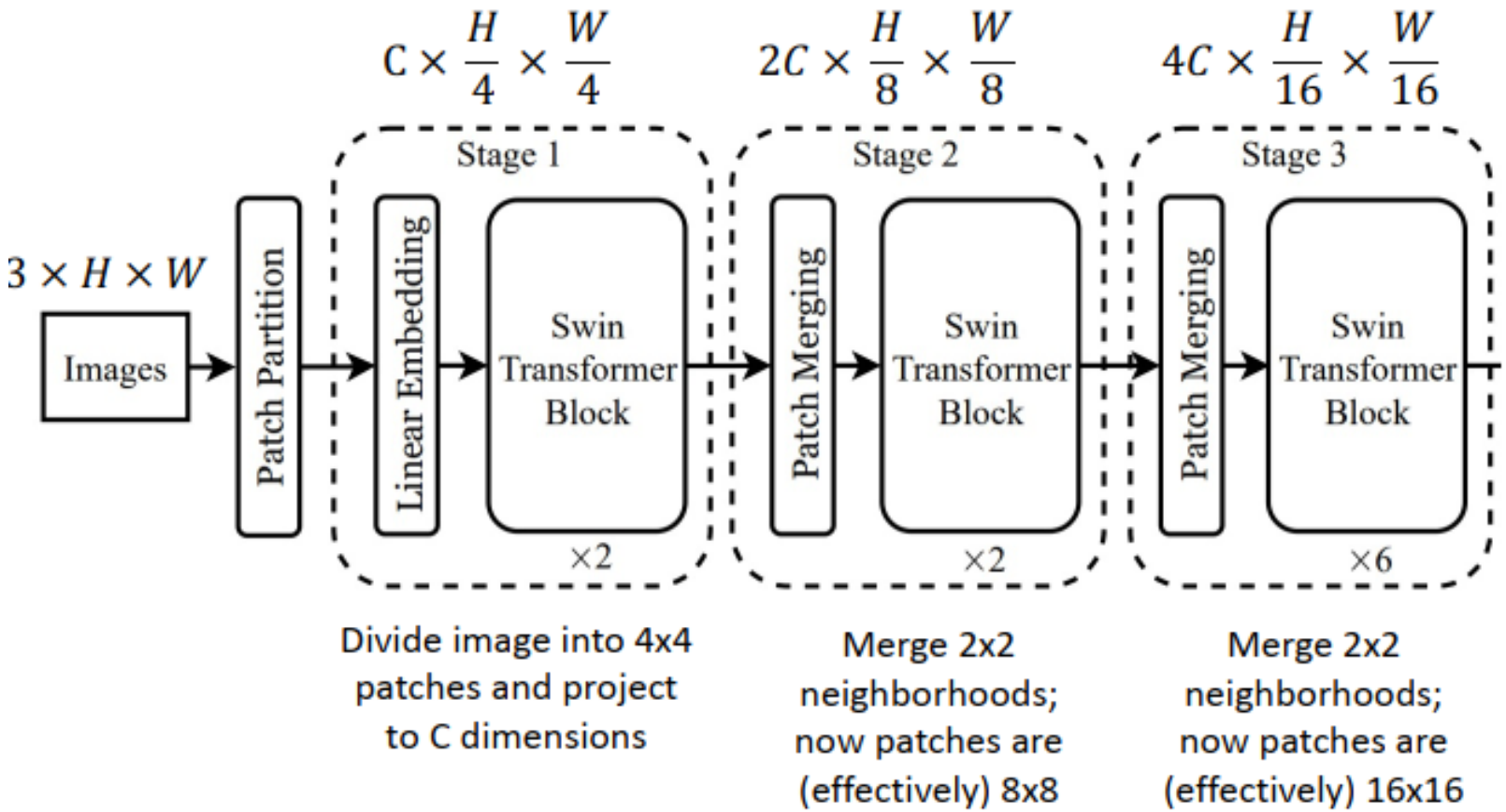


Swin Transformers

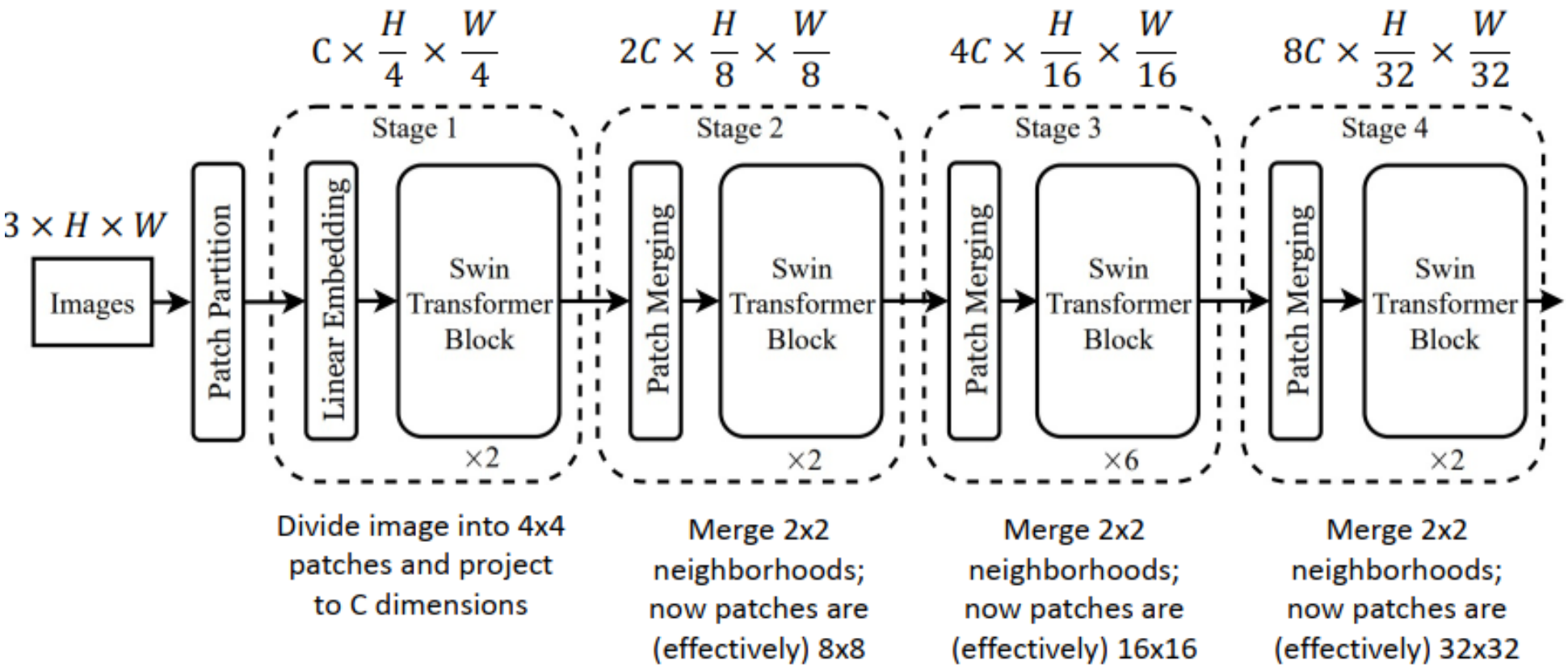




Swin Transformers

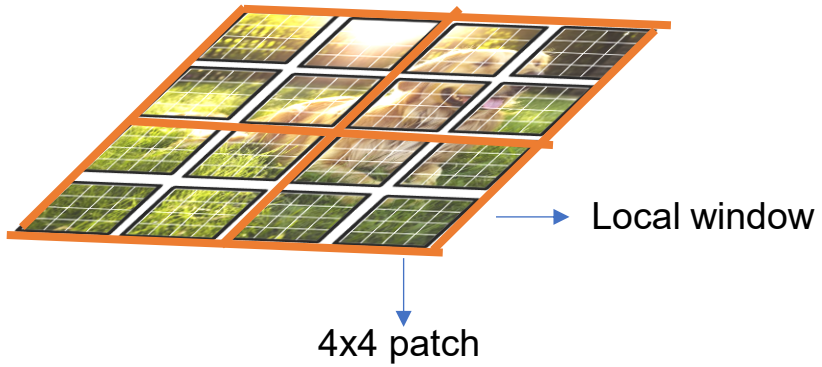


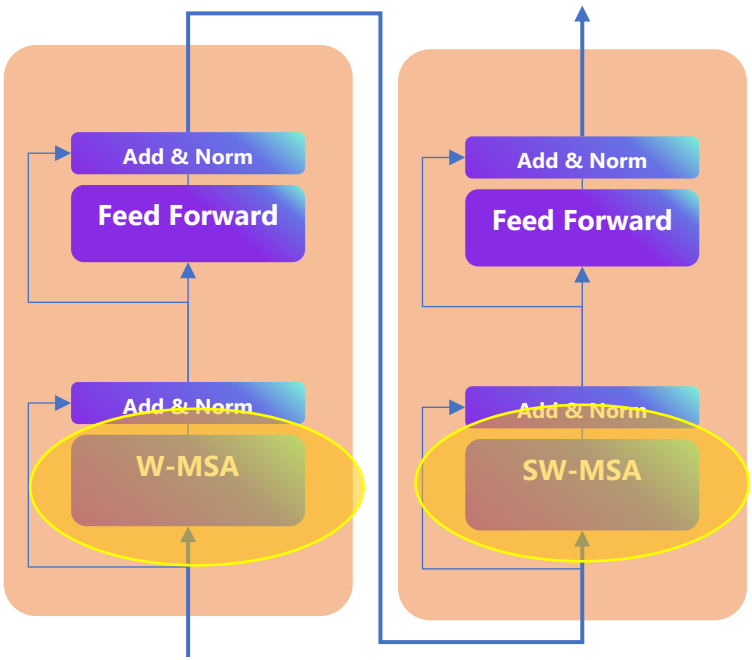
Swin Transformers

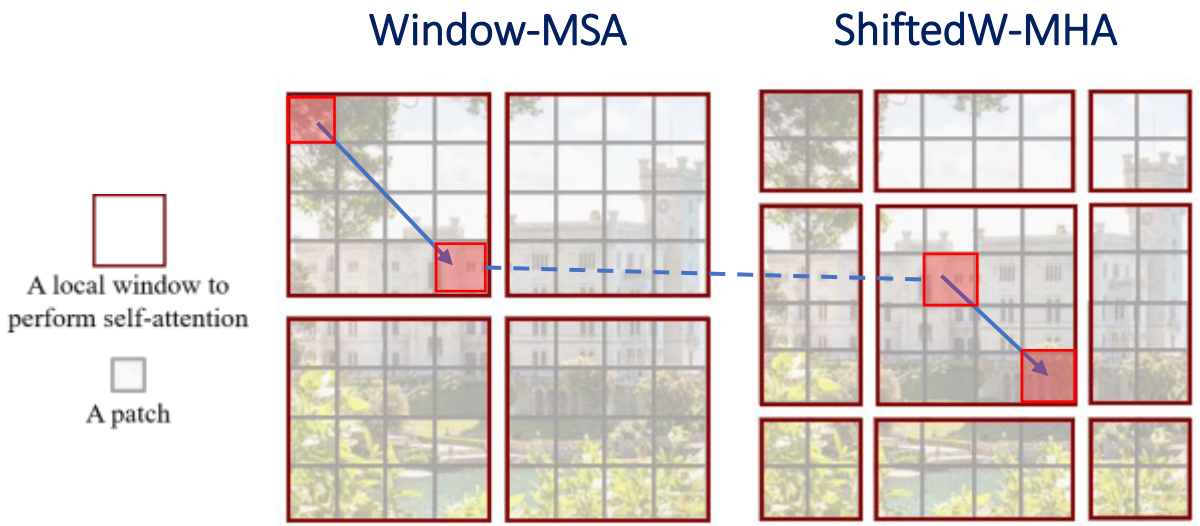


Swin Transformer Block

- ✓ Two **cascading Transformer Encoder** Blocks:
  - ✓ Window Self Attention
  - ✓ Shifted Window Self Attention



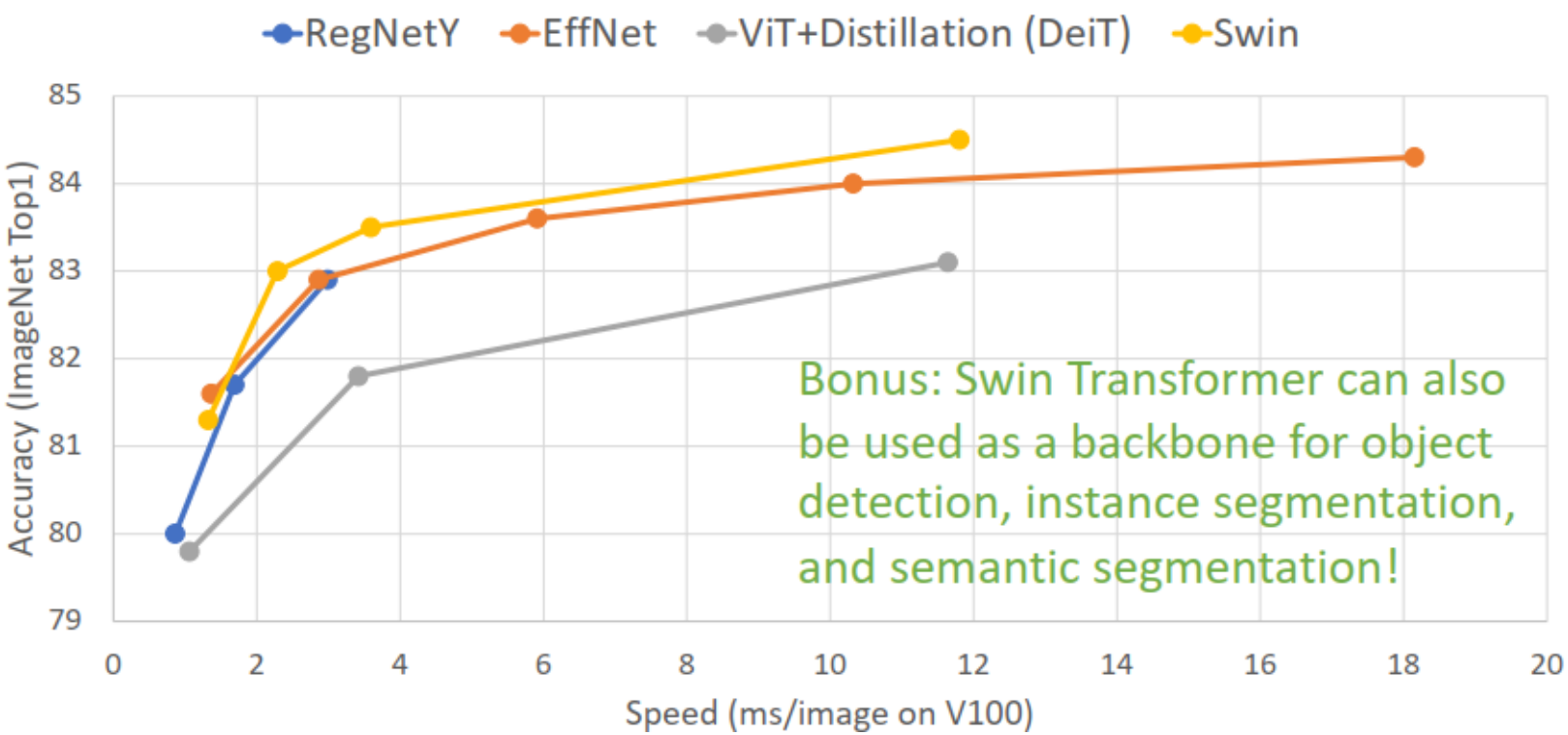




Problem: tokens only interact with other tokens within the same window; no communication across windows

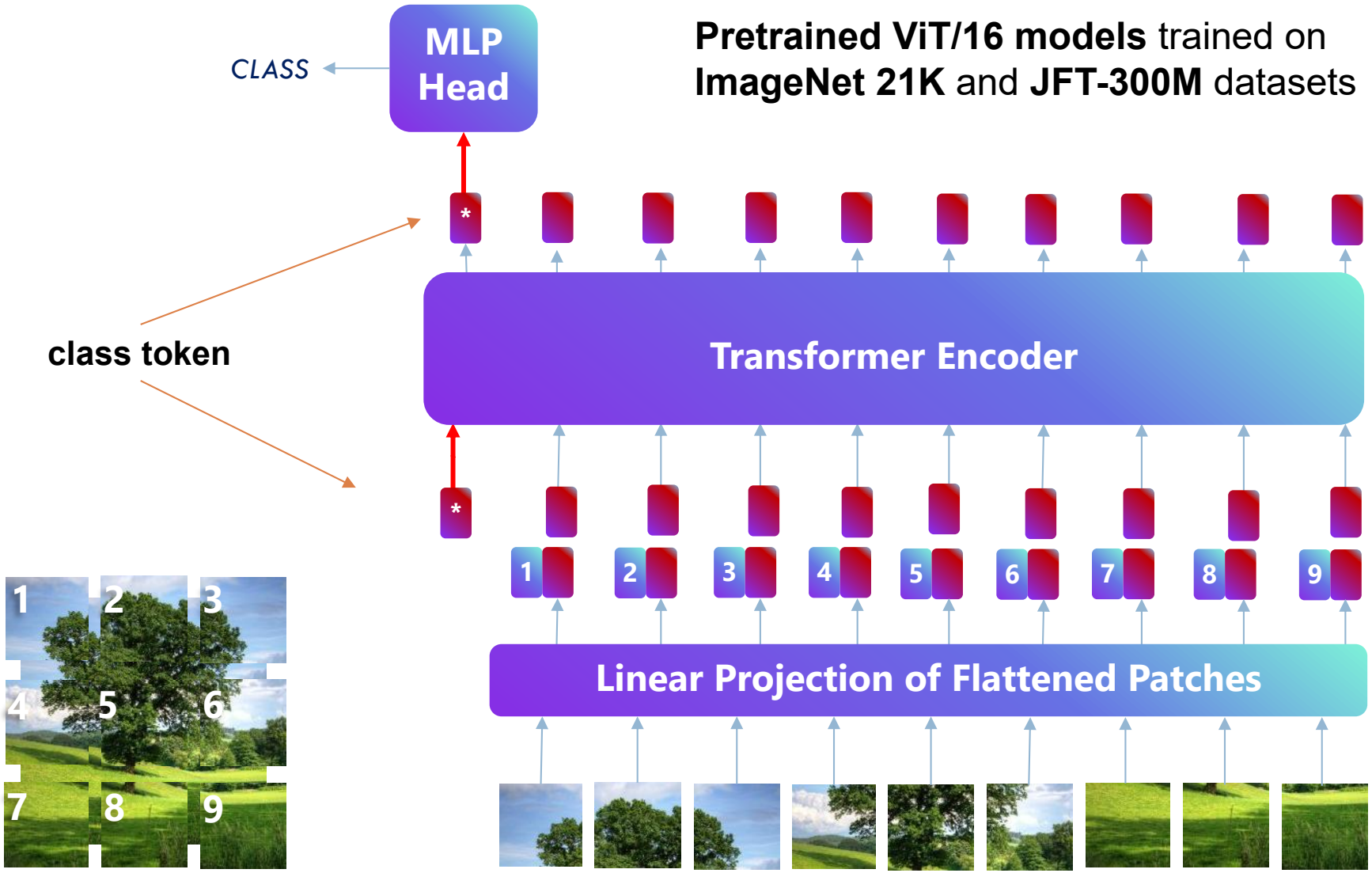
Solution: Alternate between normal windows and shifted windows in successive Transformer blocks

Swin Transformer: Speed and Accuracy

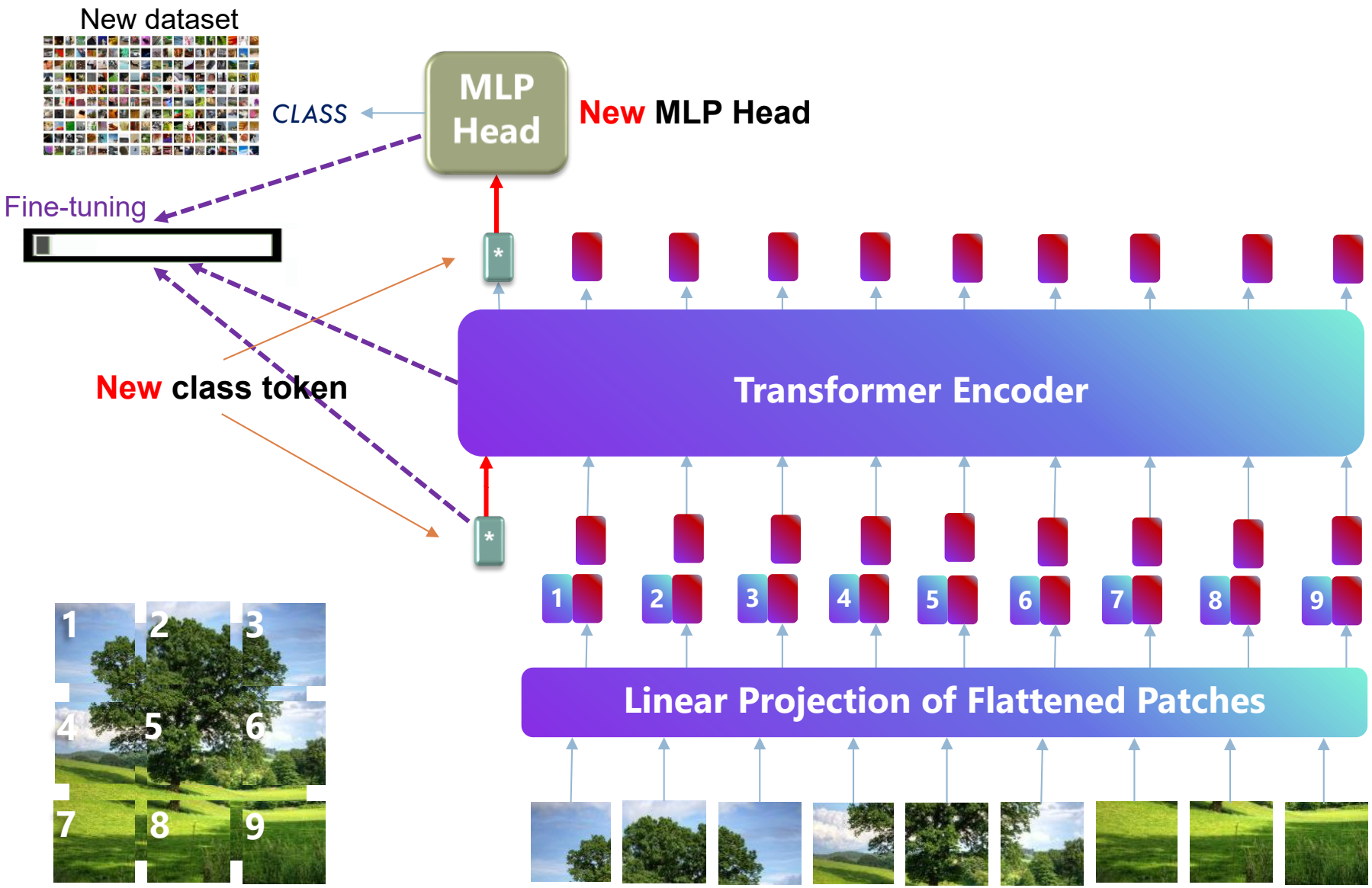


Model Fine-Tuning

ViT: Model Fine-Tuning



ViT: Model Fine-Tuning



ViT: Model Fine-Tuning with Additional Components

New dataset

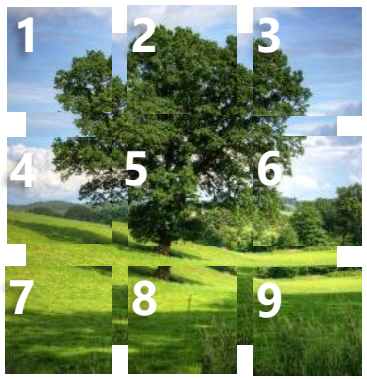


CLASS

New MLP Head

Principle: *insert to a pre-trained ViT **some additional components** that favour the computations of ViT and **fine-tune them**.*

class token



Add & Norm
Position-wise Feed Forward
Add & Norm
Multi-Head Attention
residual connection
Encoder Block
residual connection

Adapter

W_Q → LoRA
W_K → LoRA
W_V → LoRA

$X = \{x_1, x_2, x_3, x_4\}$

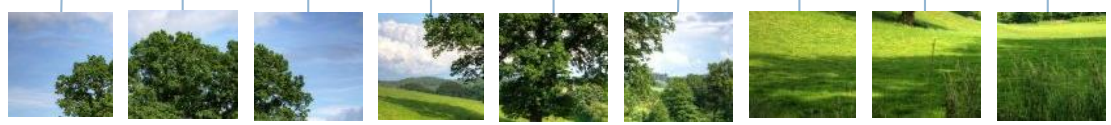
$Q = XW_Q$
$K = XW_K$
$V = XW_V$

$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$

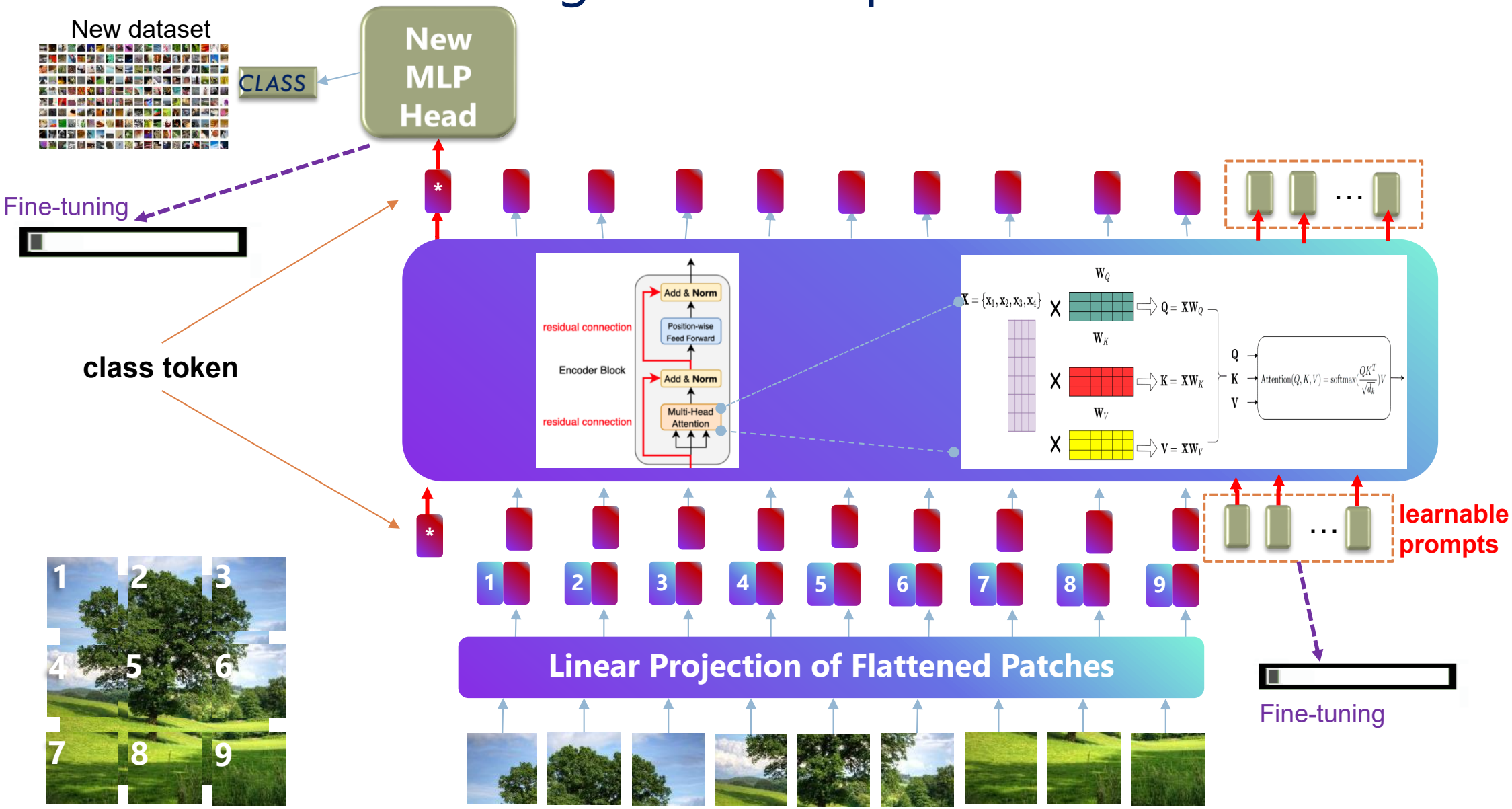
learnable prompts

1 2 3 4 5 6 7 8 9

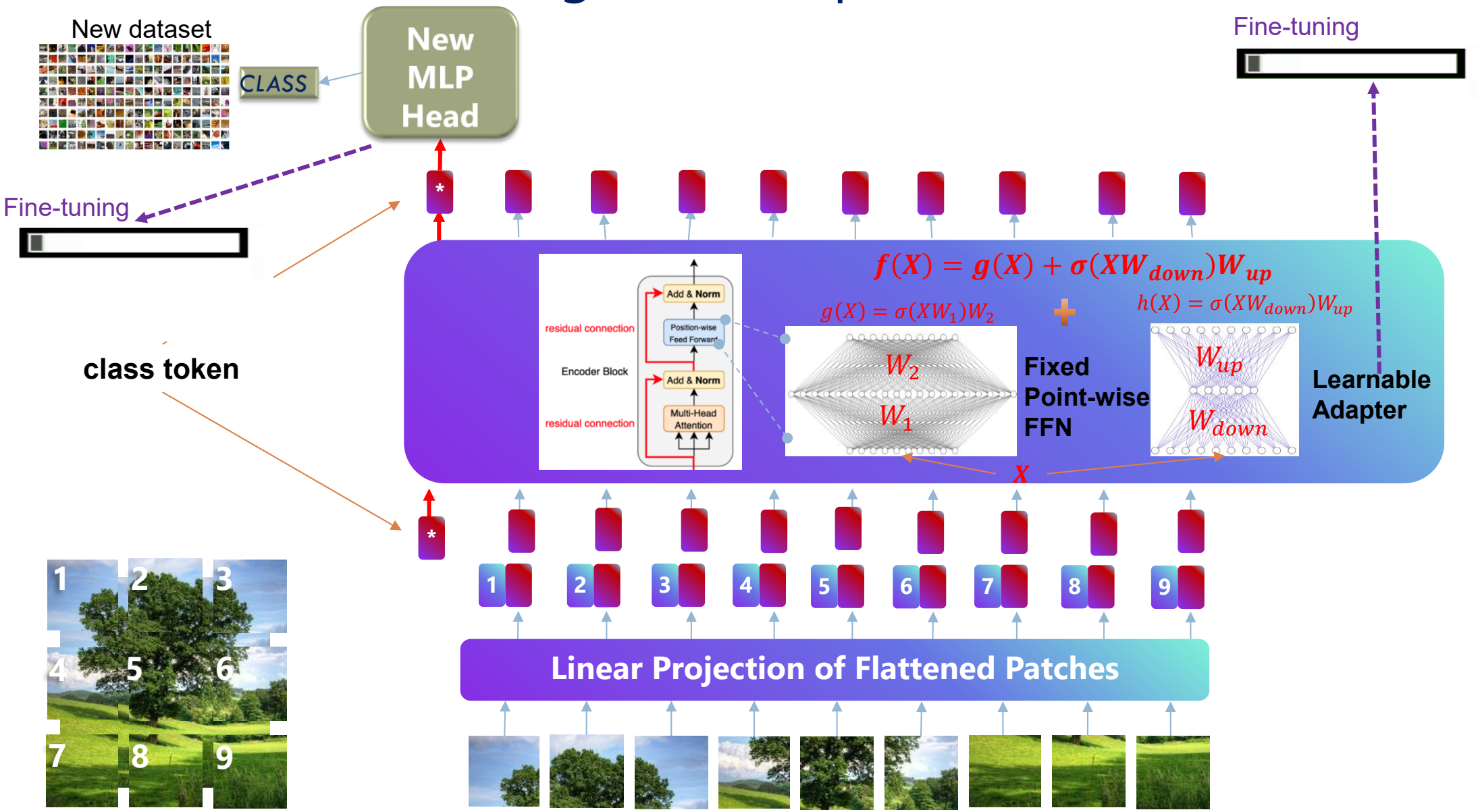
Linear Projection of Flattened Patches



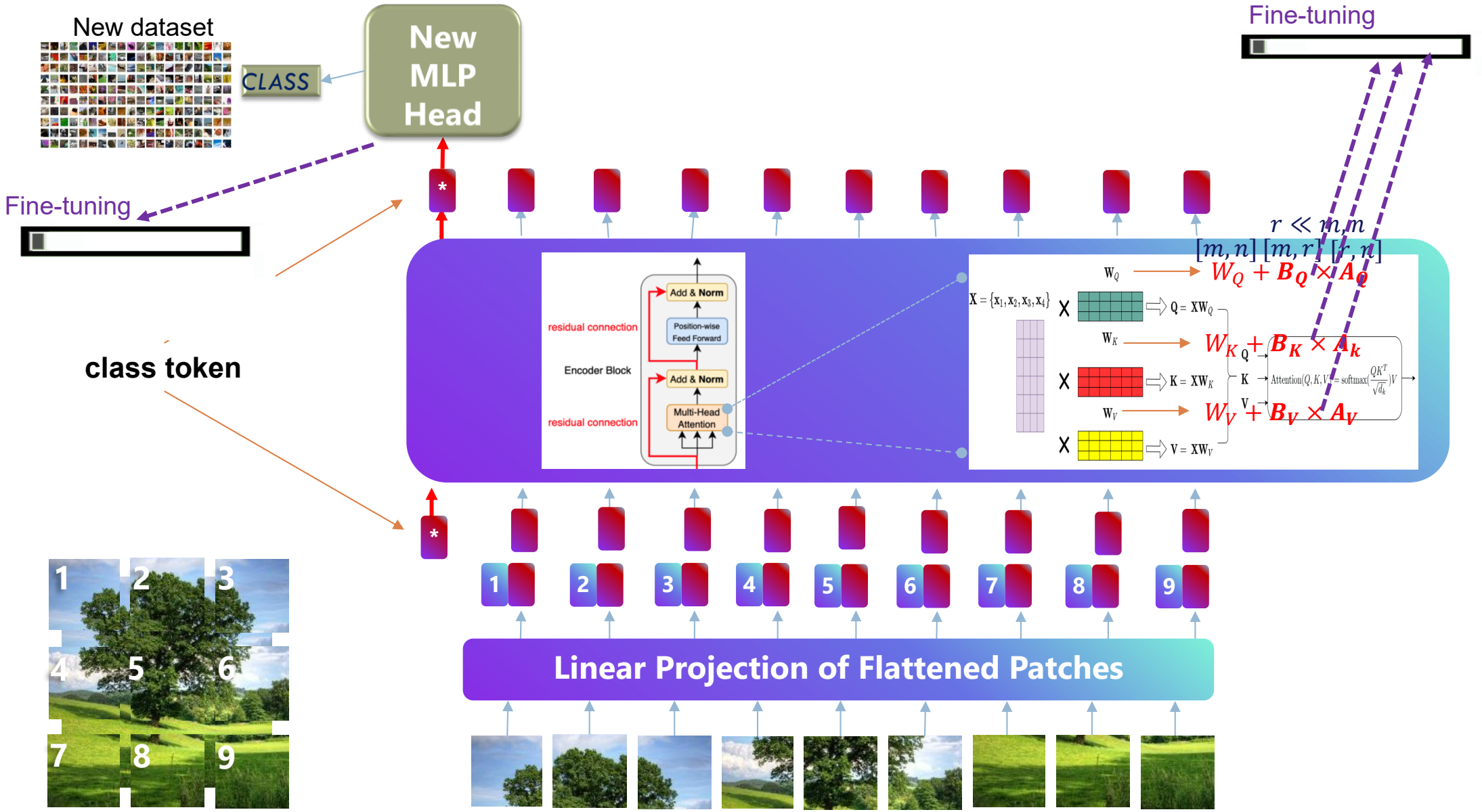
ViT: Model Fine-Tuning with Prompts



ViT: Model Fine-Tuning with Adapters



ViT: Model Fine-Tuning with Additional LoRA



Summary

- Revision of Transformers
- Vision Transformer
- Swin Vision Transformer
- Model Fine-Tuning for Transformer
  - Prompt-Tuning
  - LoRA
  - Adapter

Thanks for your attention!

Question time

